# AI Infrastructure and Regional Electricity Demand:

# Evidence from U.S. Interconnection Queues

Alan Lamb

University of Maryland

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alamb2@umd.edu

## Abstract

Between 2019 and 2025, electricity demand in ERCOT — the Texas grid — grew by roughly 27 percent. In PJM and MISO, the two largest electricity markets in the country, demand was essentially flat. This divergence coincides with the largest surge in data center investment in American history and with ERCOT interconnection queue filings that reached 97 gigawatts in 2024 alone. This paper asks how much of that divergence data center investment can explain, and whether the existing investment pipeline is large enough to stress the grid in ways that current planning models are not built to anticipate. I combine hourly electricity demand data from the EIA with interconnection queue data compiled by Lawrence Berkeley National Laboratory and the University of Maryland to build a four-layer identification strategy: panel regression, synthetic control, difference-in-differences with low-exposure controls, and narrative validation against known data center energization dates. The main empirical result is a 34.8 index-point divergence in ERCOT minimum hourly demand above a synthetic counterfactual built from balancing authorities with low data center exposure — the baseload floor signature of infrastructure that maintains high minimum power draw regardless of weather or time of day. Clean causal identification is not currently possible. The project-level large load queue data that would make it possible does not exist as a public record in any major U.S. electricity market. I document this absence as a policy finding and argue it represents a significant gap in public oversight of the grid.

**Keywords:** electricity demand, data centers, interconnection queue, synthetic control, panel regression, grid stress, energy forecasting

**JEL Codes:** Q41, Q47, L94, C23, C14

## 1. Introduction

The United States is building data centers at a pace the grid was not designed to absorb.

In 2024, Amazon, Microsoft, Google, and Meta collectively spent over $200 billion on capital expenditures, a 62 percent increase from the prior year (Mural et al., 2026). The scale of individual facilities has grown alongside the aggregate investment. Today only 10 percent of data centers draw more than 50 megawatts, but over the next decade the average new facility is projected to exceed 100 megawatts, and of the roughly 150 significant new projects announced in 2024 alone, nearly a quarter exceed 500 megawatts (BloombergNEF, 2025). Unlike industrial load, which fluctuates with economic activity, or residential load, which tracks the weather and the clock, data centers maintain a persistently high minimum power draw. Servers and cooling infrastructure run continuously, and demand does not collapse at night or on weekends the way residential and commercial load does.

This load signature shows up most clearly in ERCOT. Texas offered what data center developers were looking for: land, relatively cheap power, a deregulated market that could move quickly, and a grid authority willing to process interconnection requests at scale. By 2024, ERCOT had 97 gigawatts of new generation requests in its interconnection queue — much of it filed in anticipation of load that had not yet come online. Electricity demand in ERCOT grew by roughly 27 percent between 2019 and 2025. In PJM, which covers the Mid-Atlantic and parts of the Midwest, and MISO, which covers a large swath of the Midwest and parts of the South, demand barely moved over the same period.

The scale of the planning challenge is visible in how quickly utility forecasts have been revised. Five-year peak demand growth projections rose from 38 gigawatts in 2023 to 128 gigawatts by the end of 2024 — more than tripling in two years (Grid Strategies, 2023; Grid Strategies, 2024). Yet artificial intelligence load is not categorically tracked in any publicly available utility forecast (Grid Strategies, 2025). Without a consolidated public picture of what is coming, outside researchers, policymakers, and competing utilities cannot independently verify what grid planners are projecting or why.

What caused the ERCOT divergence is not a mystery. Measuring it precisely is harder, for a reason worth documenting carefully. Every major U.S. electricity market maintains a queue of generation interconnection requests in the form of formal, public filings made by developers who want to connect new power plants to the grid, containing precise information about location, capacity, and filing date. What the markets do not make publicly available is the equivalent record for large load customers: the queue of data center developers requesting the right to draw power from the grid at scale. PJM, ERCOT, and MISO all maintain such queues internally. None of them release the data in usable form. Requests to interconnect a large load facility are treated as proprietary, particularly for data centers where energy use is tied directly to computing power and competitive position (Goldsmith & Byrum, 2025). The result is that the three primary markets studied here — which together served approximately 61 percent of U.S. electricity load over the 2019–2025 sample period — are subject to one of the most rapid buildouts of electricity-intensive infrastructure in recent American history, and the public data needed to measure it directly does not exist.

This paper works with what is available. I use generation-side interconnection queue filings as a proxy for data center investment activity, since the generation capacity being requested exists primarily to serve the load being built. I combine these filings with hourly electricity demand from the EIA's Form 930 and with weather and economic controls to estimate the relationship between queue filing activity and subsequent demand growth across ERCOT, PJM, and MISO.

Because the identification problem is genuinely hard with three units and a proxy treatment variable, I use four methods rather than one. A panel regression gives a conditional association estimate with honest inference constraints — three clusters makes conventional significance thresholds unachievable. A synthetic control, using ERCOT as the treated unit and balancing authorities with low data center exposure as the donor pool, gives a gap estimate that is explicitly a lower bound: the donors are themselves receiving investment, which suppresses the estimated gap. A difference-in-differences specification using the cleanest available control group pushes the estimate higher. And a narrative validation overlaying known data center energization dates on the synthetic control gap plot connects the statistical finding to documented events on the ground.

The main empirical result comes from the synthetic control applied to minimum hourly demand. Minimum hourly demand captures the load that persists regardless of temperature, time, or season. Using a donor pool of balancing authorities with low data center exposure, the analysis produces a gap of 34.8 index points — meaning ERCOT's minimum demand was 34.8 percent above the level the synthetic counterfactual predicts it would have reached without the data center surge — with zero of three placebo units exceeding ERCOT's divergence.

The paper proceeds as follows. Section 2 reviews the relevant literature. Section 3 describes the data, including an extended discussion of what does not exist and what its absence costs. Section 4 presents descriptive evidence. Section 5 lays out the empirical strategy before any results are shown. Section 6 presents results across all four identification layers. Section 7 synthesizes the estimated range and presents the CAISO comparative analysis. Section 8 describes the forecasting model. Section 9 addresses policy implications. Section 10 concludes.

## 2. Related Literature

### 2.1 Interconnection Queue and Grid Infrastructure

The academic literature on interconnection queues has focused almost entirely on the supply side, including the costs, delays, and withdrawal rates associated with new generation projects seeking grid access. Johnston, Liu, and Yang (2023) provide the most comprehensive empirical treatment in the academic literature, documenting the dramatic increase in queue volumes, high interconnection costs as a primary driver of project withdrawal, and the substantial network upgrade costs that fall on new entrants. The LBNL annual Queued Up analysis documents that only 13 to 19 percent of projects entering the queue ultimately reach commercial operation, implying withdrawal rates above 80 percent in recent cohorts (Rand et al., 2024).

This paper extends that supply-side literature to the demand side. Where Johnston, Liu, and Yang ask what happens to generation projects in the queue, I ask what the queue predicts about subsequent electricity consumption. The demand-side interconnection queue, including the list of large load customers seeking to connect at scale is estimated to represent hundreds of gigawatts of data center projects in the United States alone. Yet this data is managed by utilities and is not included in any publicly available dataset (BCG, 2025). To my knowledge, no existing paper uses generation interconnection queue filings as a leading indicator of realized regional electricity demand.

The institutional framework governing interconnection queues has evolved through a series of Federal Energy Regulatory Commission rulemakings. FERC Order 2000 established regional transmission organizations and required that the generation interconnection queue be made publicly available, thus creating the data infrastructure on which this paper relies (FERC, 1999). Order 890 in 2007 strengthened open-access requirements and standardized interconnection procedures. Order 2023 in 2023 implemented the most significant reform to the interconnection process in two decades, moving from a serial first-come, first-served study process to a cluster-based approach. The transition to Order 2023 produced a wave of batch filings as developers rushed to secure positions under the prior rules — an institutional artifact that appears in the MISO queue series and is addressed in the data section.

### 2.2 Data Center Energy Consumption

The Lawrence Berkeley National Laboratory has produced the most authoritative estimates of U.S. data center energy consumption. Their 2024 report documents the acceleration in data center power draw since 2022, driven by AI-intensive workloads that require substantially more compute — and therefore more power — than traditional cloud applications. U.S. data centers consumed approximately 183 terawatt-hours of electricity in 2024, equivalent to more than four percent of total national consumption (IEA, 2025). By 2030, the IEA projects this figure will grow by approximately 130 percent.

The IEA's Energy and AI report projects that global data center consumption will reach 945 terawatt-hours annually by 2030, more than double the 2024 figure (IEA, 2025). The U.S. and China together account for nearly 80 percent of projected global growth. The concentration of this load in specific regions — Virginia, Texas, Ohio — creates the kind of geographically clustered demand shock that aggregate national estimates cannot capture, and that this paper is designed to measure.

Data centers differ from most electricity consumers in their load profile. Unlike residential or commercial load, which tracks the time of day and weather, data centers operate continuously and maintain a persistently high minimum draw. Knittel et al. (2025) model data center temporal flexibility across three major U.S. regions and find that even under high flexibility assumptions — where facilities shift workloads between hours to reduce peak demand — data centers retain a substantial baseline power requirement that does not approach zero. This load profile has important implications for how the demand signature of data center growth appears in regional electricity data, and motivates the minimum demand specification developed in Section 5.

### 2.3 Electricity Demand Forecasting

Electricity demand forecasting has a long methodological tradition in energy economics. Weron (2014) provides a comprehensive review of forecasting methods across time series, econometric, and machine learning approaches. The consensus finding is that no single method dominates across all contexts: pure time series models perform well at short horizons where recent patterns are informative, while structural models incorporating economic and demographic drivers perform better over longer horizons where secular trends matter.

The academic literature on regional electricity demand has relied primarily on panel data methods to identify the drivers of consumption growth. Studies in this tradition typically regress electricity demand on income, price, weather, and industrial composition to identify elasticities. The challenge this paper confronts is identifying a specific new demand source that arrives gradually and is correlated with underlying economic conditions. This is substantially harder than estimating aggregate income elasticities, and the literature offers limited methodological guidance.

A related dimension of this forecasting challenge concerns load factor, measured as the ratio of average demand to peak demand. Hledik, Lam, and Peters (2026) model large transmission-connected loads such as data centers at a 70 percent annual load factor, substantially above the 40 to 50 percent load factors typical of residential and commercial customers. This distinction matters for grid planning: high load factor customers add more to the energy base than to the system peak, and their impact is therefore more visible in measures of minimum and average demand than in peak demand alone. Brattle (2024) further distinguishes between large discrete loads that arrive quickly, carry high uncertainty, and can be built within months, and the smoother load growth from electrification and efficiency that traditional utility forecasts were designed to handle. This distinction motivates the minimum demand specification developed in Section 5. Discrete always-on load shows up in the demand floor in a way that aggregate forecasting methods are not designed to detect.

The demand implications of data center load growth have received increasing attention from applied researchers. Carnegie Mellon University researchers have estimated that data centers and cryptocurrency mining could increase average U.S. electricity bills by 8 percent by 2030, with increases exceeding 25 percent in the highest-demand markets of central and northern Virginia (Carnegie Mellon University/North Carolina State University, 2024).

### 2.4 Synthetic Control Methods

The synthetic control method was introduced by Abadie, Diamond, and Hainmueller (2010) and extended by Abadie (2021). I follow the recommendations in Abadie (2021) on placebo inference, pre-treatment fit assessment, and the interpretation of results when the donor pool is small. PJM and MISO both received real data center investment during the sample period, creating a contaminated donor problem. This is acknowledged explicitly and used to frame the synthetic control results as a lower bound rather than a point estimate, consistent with the guidance in Abadie (2021) on settings where no clean control group exists.

## 3. Data

### 3.1 EIA Form 930 — Hourly Electricity Demand

The primary outcome variable is regional electricity demand, drawn from the Energy Information Administration's Form 930 — the Hourly Electric Grid Monitor. This dataset reports hourly demand in megawatt-hours for each balancing authority in the continental United States, updated in near real-time and publicly available through the EIA's open data API.

I pull all available demand observations for three balancing authorities — ERCO (ERCOT), PJM, and MISO — covering January 2019 through December 2025. After filtering to demand-type observations only and removing three corrupt rows from October 2021 that reported values in the billions of megawatt-hours, the raw dataset contains 184,102 hourly observations. These are aggregated to monthly averages to construct the unit of observation for the panel: one row per balancing authority per calendar month, yielding 252 observations across 84 months.

Two outcome variables are constructed from this series. The first is average hourly demand: the mean of all hourly readings within a given BA-month. The second is minimum hourly demand: the lowest hourly reading within a given BA-month. This second measure is the basis for the headline synthetic control result. Minimum demand isolates the baseload floor — the load that persists regardless of weather, time of day, or season. As documented in Section 2, data centers maintain a persistently high minimum power draw even under flexible operating conditions (Knittel et al., 2025; Hledik, Lam, and Peters, 2026). This load profile should appear more clearly in minimum demand than in average demand, which is heavily influenced by residential and commercial consumption patterns that fluctuate with temperature.

### 3.2 Interconnection Queue Data

The treatment variable is constructed from two interconnection queue datasets.

The first is the Lawrence Berkeley National Laboratory's Queued Up dataset (2024 edition), which compiles interconnection queue filings across all major U.S. electricity markets from the late 1990s through the end of 2024. The full dataset covers projects nationwide; after filtering to the three primary balancing authorities — PJM, MISO, and ERCOT — and removing records with missing capacity values or invalid filing dates, the working dataset for the primary analysis contains 16,093 project-level observations. Each row represents a single interconnection request and includes the filing date, project capacity in megawatts, fuel type, project status, and the balancing authority in which the project is located. An expanded version of the dataset, retaining three additional low-exposure balancing authorities for use as donor and control units in the synthetic control and difference-in-differences specifications, is described in Section 5.

The second is the University of Maryland interconnection queue dataset compiled by Yang, Johnston, and Liu (NBER Working Paper 31946), which provides enhanced project-level data for PJM covering 2008 through 2024. This dataset includes 7,492 observations across 121 variables, with geographically precise location data and detailed cost information not available in the LBNL data.

From these sources I construct two queue variables. The flow variable, queue_mw_filed, measures the total megawatts of new interconnection requests filed within a given BA-month. The stock variable, queue_mw_active, measures the cumulative megawatts ever filed in each BA through a given month. A separate variable, queue_mw_filed_large, restricts the flow measure to projects at or above 100 megawatts, the threshold used to identify hyperscale-class load based on the project size distribution described in Section 3.2.1.

### 3.2.1 MW Threshold Selection

The 100 megawatt threshold for large projects was determined empirically by examining the distribution of project sizes in the LBNL queue data. The distribution is smooth and right-skewed with no visible natural break, which is consistent with the absence of a single dominant project type. The 100 megawatt threshold was selected based on its correspondence with the industry definition of hyperscale infrastructure (BloombergNEF, 2025) and validated through sensitivity checks at 50 and 200 megawatts. Results are stable across all three thresholds and are reported in full in the robustness table in Section 6.

### 3.3 Weather Controls

Electricity demand has strong seasonal patterns driven by heating and cooling load, and failure to control for weather would confound the queue signal with temperature variation. ERCOT's hot summers produce average hourly demand that is 30 to 40 percent above mild-weather months — variation that is entirely unrelated to data center investment. I control for this using monthly heating degree days and cooling degree days by balancing authority, drawn from NOAA's National Centers for Environmental Information. Heating degree days measure the cumulative departure of daily average temperature below 65 degrees Fahrenheit within a month; cooling degree days measure the equivalent departure above 65 degrees. Together they capture the primary weather-driven component of electricity demand and allow the queue variable to be identified from variation that remains after removing seasonal temperature effects.

### 3.4 Economic Controls

Regional economic activity is a determinant of electricity demand independent of data center investment. Faster-growing regional economies consume more electricity through industrial output, commercial activity, and population-driven residential demand. Without controlling for this, a positive relationship between queue filings and demand growth could reflect underlying regional economic conditions that independently drive both. I control for annual real GDP by state, drawn from the Bureau of Economic Analysis regional accounts and assigned to balancing authorities by geographic crosswalk. The GDP control absorbs secular economic differences across balancing authorities over time. Its exclusion is tested in Panel C of the robustness table in Section 6; the primary coefficient is stable with and without it.

### 3.5 The Data Gap

The generation interconnection queue is a public record. PJM, ERCOT, MISO, and the Lawrence Berkeley National Laboratory all publish queue filings in searchable, downloadable form — which is why this paper can use them as a treatment variable at all. FERC Order 2000 required this transparency, and it created the data infrastructure on which the present analysis depends.

No equivalent transparency requirement exists for the load side. Every major U.S. electricity market maintains an internal queue of large load interconnection requests — filings made by data center operators, industrial customers, and other large electricity consumers seeking to connect to the grid at scale. This data is not publicly available in any of the three primary markets studied here. Requests to interconnect a large load facility are treated as proprietary, particularly for data centers where energy use is tied directly to computing power and competitive position (Goldsmith & Byrum, 2025).

The practical consequence for research is direct. With facility-level load queue data, the identification problem this paper addresses by proxy — using generation queue filings as a leading indicator of load growth — could be addressed directly. Causal estimates of the per-facility demand effect would be achievable. The proxy variable would become unnecessary. Planning models could incorporate the actual load pipeline rather than inferring it from the generation side.

The practical consequence for grid planning is larger. Utility five-year peak demand forecasts more than tripled between 2023 and 2024 — from 38 gigawatts to 128 gigawatts — without access to the actual load pipeline (Grid Strategies, 2023; Grid Strategies, 2024). Artificial intelligence load is not categorically tracked in any publicly available utility forecast (Grid Strategies, 2025). Planners are building infrastructure that will last decades in the absence of direct observation of what is coming.

The three balancing authorities studied here served approximately 61 percent of U.S. electricity load over the 2019–2025 sample period. They are also the regions experiencing the most concentrated data center investment in the country. The gap in public data infrastructure is therefore not a methodological inconvenience specific to this paper, it is an absence of oversight at a moment when the scale and pace of investment make oversight most consequential.

Two jurisdictions have moved toward greater transparency without waiting for a federal mandate. Since the first quarter of 2024, Georgia Power has published quarterly Large Load Economic Development Reports under its Integrated Resource Plan, publicly disclosing the size, review status, proposed service date, and load ramp schedule of all commercial large load projects above 115 megawatts (Goldsmith & Byrum, 2025). Texas, through Senate Bill 6 in 2025, required data centers above 75 megawatts to submit formal load interconnection requests with associated disclosure requirements (Texas SB-6, 2025). These are meaningful precedents. They demonstrate that structured large-load disclosure is administratively achievable and does not require revealing commercially sensitive information beyond what generation developers already disclose under Order 2000 — project capacity, proposed location, and expected service date.

The case for broader disclosure is not that companies should be required to reveal proprietary competitive strategy. It is that the scale, location, and timing of infrastructure drawing power from a shared public grid is information with significant public consequences. Generation developers making equivalent commitments have disclosed this information for decades under federal requirements. The asymmetry between generation-side and load-side transparency has no technical justification; it reflects a regulatory framework built before large load customers became material actors in grid planning. Whether FERC should require equivalent load-side disclosure through rulemaking, whether RTOs should adopt it voluntarily, and how confidentiality protections should be structured are questions beyond the scope of this paper. What this paper can document is the cost of the current arrangement.

That cost can be stated with some precision. The forecast confidence intervals that Section 8 will produce are expected to be wider than they would be with facility-level load data, because the model must infer the demand pipeline from generation-side filings rather than observing it directly. The gap between the ARIMA baseline forecast and the XGBoost model incorporating queue features will be a direct measure of the informational value of the generation proxy. The remaining uncertainty in the XGBoost forecast, the portion that better load-side data would reduce, will quantify the cost of the current disclosure gap. That number will not be a statistical artifact. It will measure, in concrete terms, what is being lost

## 4. Descriptive Evidence

Table 1 presents descriptive statistics for the panel by balancing authority. Several features of the data warrant attention before turning to the regression results. ERCOT is the smallest market by average hourly demand (48,634 MWh), roughly half the size of PJM (91,586 MWh) and smaller than MISO (73,653 MWh), yet its average monthly large-project queue filings (4,569 MW) exceed PJM's (2,831 MW) on a flow basis — a disproportionately high level of filing activity relative to market size that reflects the surge in ERCOT interconnection requests documented in Section 1. The ratio of minimum to average monthly demand — the demand floor as a share of average consumption — is 72.8 percent in ERCOT, 74.7 percent in PJM, and 76.8 percent in MISO. This ratio matters for interpreting the synthetic control results in Section 6.3: because ERCOT's demand floor is relatively more weather-sensitive than PJM's or MISO's, any rise in that floor that cannot be explained by temperature is a stronger signal of always-on infrastructure load. Weather patterns are coherent with geography: MISO records the highest average heating degree days (469), reflecting its upper Midwest coverage, while ERCOT records the highest average cooling degree days (263), consistent with Texas summers. The large standard deviation in MISO queue filings (20,989 MW against a mean of 5,334 MW) reflects a batch-filing artifact from the 2022 queue reform; this is addressed in the winsorization note in Appendix C.

**Table 1. Descriptive Statistics**

| Variable | N | Mean | Std. Dev. | Min | Max |
|---|---|---|---|---|---|
| **Panel A: ERCOT** | | | | | |
| Avg. hourly demand (MWh) | 84 | 48,634 | 8,211 | 36,647 | 67,863 |
| Min. hourly demand (MWh) | 84 | 35,374 | 5,394 | 27,449 | 49,402 |
| Queue MW filed, ≥100 MW | 84 | 4,569 | 3,271 | 0 | 14,974 |
| Queue MW active (stock) | 84 | 200,264 | 135,596 | 4,002 | 397,859 |
| Heating degree days | 84 | 164 | 206 | 0 | 686 |
| Cooling degree days | 84 | 263 | 264 | 0 | 864 |
| Regional GDP (billions $) | 84 | 2,007.4 | 174.7 | 1,782.7 | 2,221.9 |
| **Panel B: PJM** | | | | | |
| Avg. hourly demand (MWh) | 84 | 91,586 | 10,075 | 73,267 | 116,275 |
| Min. hourly demand (MWh) | 84 | 68,379 | 5,582 | 56,260 | 82,130 |

| | | | | | |
|---|---|---|---|---|---|
| Queue MW filed, ≥100 MW | 84 | 2,831 | 6,406 | 0 | 32,486 |
| Queue MW active (stock) | 84 | 203,997 | 98,894 | 2,504 | 294,758 |
| Heating degree days | 84 | 338 | 337 | 0 | 1,074 |
| Cooling degree days | 84 | 128 | 173 | 0 | 531 |
| Regional GDP (billions $) | 84 | 6,236.4 | 259.7 | 5,807.7 | 6,538.7 |
| **Panel C: MISO** | | | | | |
| Avg. hourly demand (MWh) | 84 | 73,653 | 7,018 | 59,987 | 90,555 |
| Min. hourly demand (MWh) | 84 | 56,577 | 3,705 | 48,878 | 66,762 |
| Queue MW filed, ≥100 MW | 84 | 5,334 | 20,989 | 0 | 154,214 |
| Queue MW active (stock) | 84 | 167,669 | 96,064 | 200 | 293,941 |
| Heating degree days | 84 | 469 | 436 | 0 | 1,377 |
| Cooling degree days | 84 | 96 | 130 | 0 | 441 |
| Regional GDP (billions $) | 84 | 2,859.7 | 106.2 | 2,671.3 | 2,980.2 |
| **Panel D: All Balancing Authorities** | | | | | |
| Avg. hourly demand (MWh) | 252 | 71,291 | 19,587 | 36,647 | 116,275 |
| Min. hourly demand (MWh) | 252 | 53,443 | 14,549 | 27,449 | 82,130 |
| Queue MW filed, ≥100 MW | 252 | 4,245 | 12,802 | 0 | 154,214 |
| Queue MW active (stock) | 252 | 190,643 | 112,396 | 200 | 397,859 |
| Heating degree days | 252 | 324 | 361 | 0 | 1,377 |
| Cooling degree days | 252 | 162 | 209 | 0 | 864 |
| Regional GDP (billions $) | 252 | 3,701.1 | 1,839.6 | 1,782.7 | 6,538.7 |

*Notes: Sample period January 2019–December 2025. Unit of observation is balancing authority × calendar month (84 months per BA; 252 total observations). Avg. hourly demand is the mean of all hourly EIA Form 930 readings within a BA-month. Min. hourly demand is the lowest single hourly reading within a BA-month. Queue MW filed (≥100 MW) is total megawatts of new interconnection requests filed by projects at or above 100 MW in a given BA-month, from LBNL Queued Up (2024 edition). Queue MW active (stock) is the cumulative megawatts ever filed through a given month. HDD and CDD from NOAA NCEI. Regional GDP is annual real GDP by state from BEA regional accounts (2017 dollars), assigned to balancing authorities by geographic crosswalk. The large standard deviation in MISO queue filings reflects a batch-filing spike in 2022 associated with queue reform dynamics rather than underlying demand pressure; see Appendix C.*

Figure 1 presents indexed average hourly electricity demand for each of the three balancing authorities from January 2019 through December 2025, normalized to 100 at the start of the sample period. The divergence is striking. ERCOT demand reached an index value of approximately 127 by the end of the sample — a 27 percent increase over six years. PJM and MISO ended the period near 102 to 105, essentially flat. This divergence is the central descriptive fact that motivates the paper. The two largest electricity markets in the United States, serving the industrial Midwest and the tech-dense Mid-Atlantic, experienced essentially no demand growth over a period in which the Texas grid grew by more than a quarter.

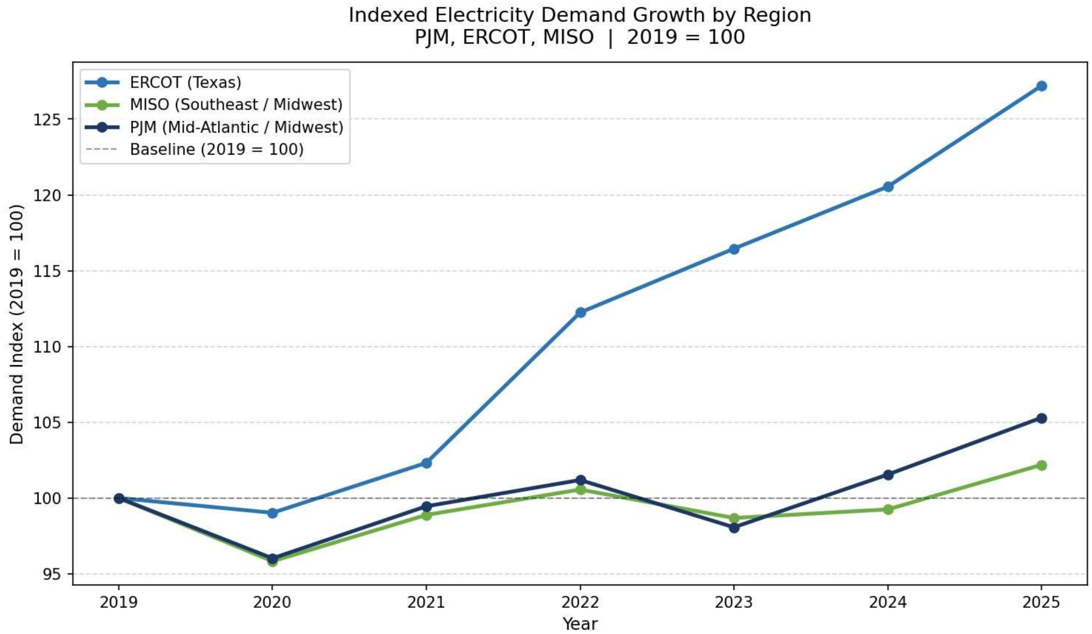



**Figure 1.** Indexed Average Hourly Electricity Demand by Region, 2019 — 2025.

*Notes: Demand indexed to 2019 annual average = 100. ERCOT (Texas), PJM (Mid-Atlantic/Midwest), MISO (Southeast/Midwest). Source: EIA Form 930; authors' calculations.*

Figure 2 presents the flow of new interconnection queue filings by balancing authority, measured in total megawatts filed per month. The ERCOT series rises sharply after 2021 and reaches a peak in 2024 that, summed annually, exceeds 97 gigawatts (LBNL Queued Up, 2024 edition). The PJM series shows elevated filings through 2022 before dropping sharply — an artifact of PJM's decision to pause new interconnection requests while it implemented reforms to its study process. The MISO series shows a pronounced spike in 2022, also attributable to queue reform dynamics rather than underlying demand pressure: developers rushed to file before new procedural requirements took effect. These institutional artifacts in the PJM and MISO series reinforce the value of using lagged specifications and robustness checks rather than relying on contemporaneous correlations.

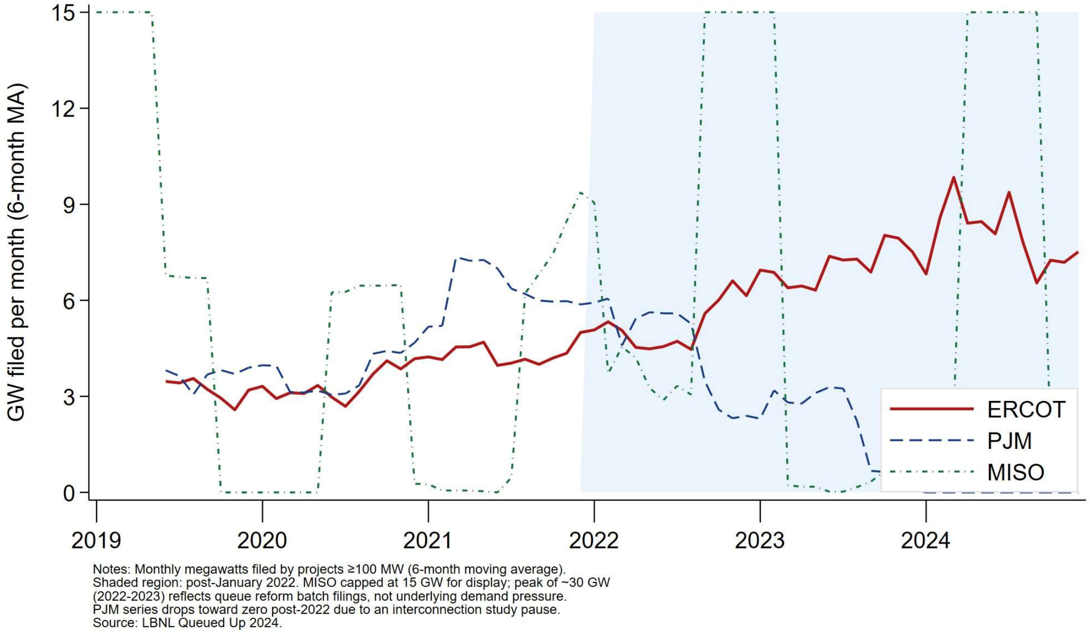



**Figure 2.** Monthly Interconnection Queue Filings by Balancing Authority, 2019 — 2024.

*Notes: Monthly megawatts filed by projects ≥100 MW (6-month moving average). Shaded region: post-January 2022. MISO capped at 15 GW for display; peak of ~30 GW (2022 — 2023) reflects queue reform batch filings, not underlying demand pressure. PJM series drops toward zero post-2022 due to an interconnection study pause. Source: LBNL Queued Up 2024.*

Figure 3 presents dual y-axis co-movement charts for each balancing authority, plotting average hourly demand against lagged queue filings on a common time axis. The ERCOT chart shows clear co-movement beginning around 2022: queue filings surge in 2021 and 2022, and average hourly demand begins climbing 18 to 24 months later — consistent with the construction timeline assumed in the panel regression. The PJM and MISO charts show no comparable co-movement, which is expected given the flatter demand trajectories and the institutional disruptions to their queue series discussed above.

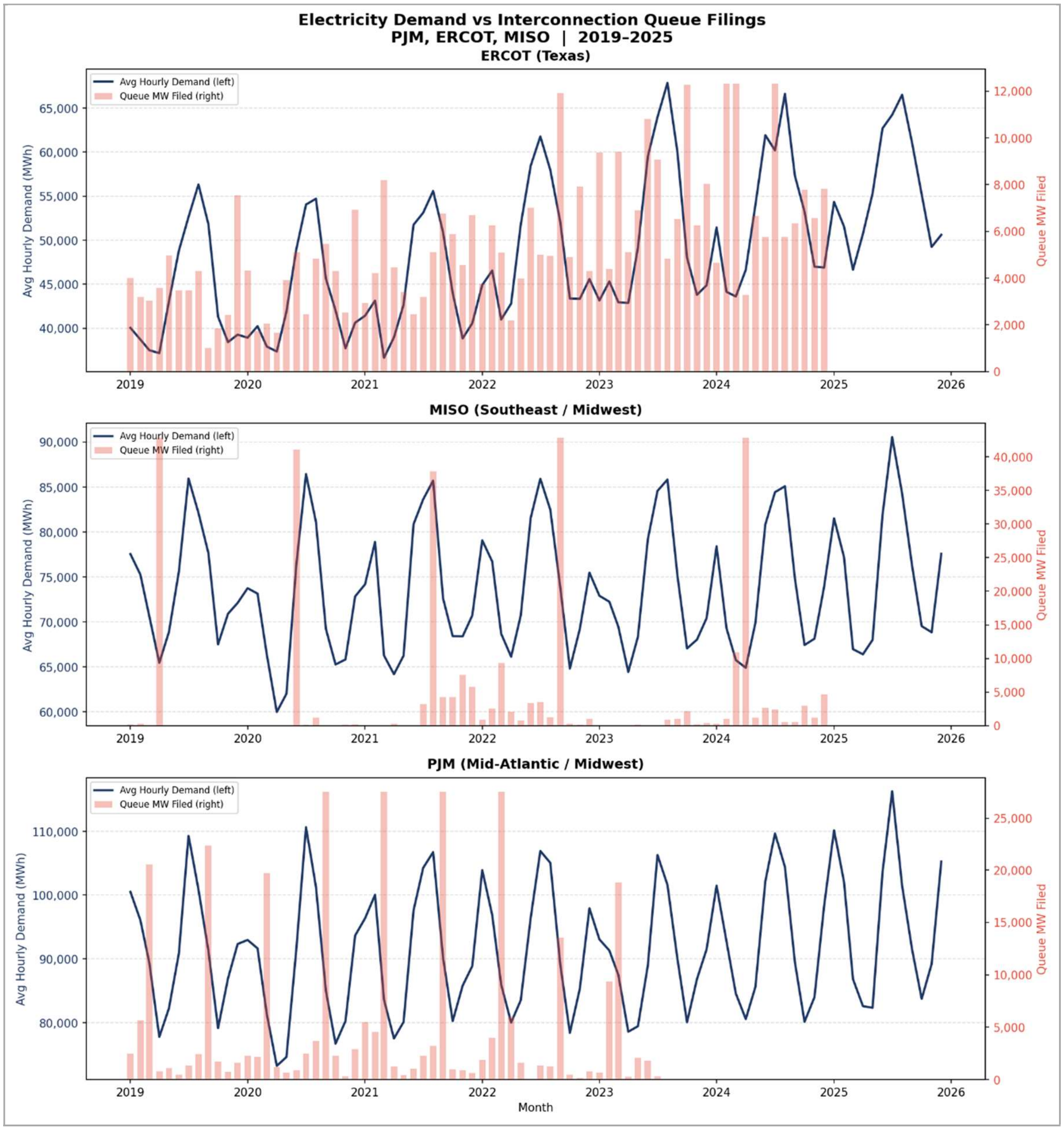



Figure 4 presents load shape evidence for ERCOT, plotting both average and minimum hourly demand on an indexed basis. Minimum demand — the lowest hourly reading in each month, which typically occurs in the early morning hours of mild-weather months — rises faster than average demand over the sample period. By 2025, ERCOT minimum demand had increased by more than average demand on an indexed basis, a pattern inconsistent with weather-driven load growth, which would affect average and minimum demand approximately proportionally. The divergence between average and minimum demand growth is the visual motivation for the minimum demand synthetic control specification. Infrastructure that draws constant power regardless of temperature or time of day — the load profile of data centers — would appear precisely in this measure.

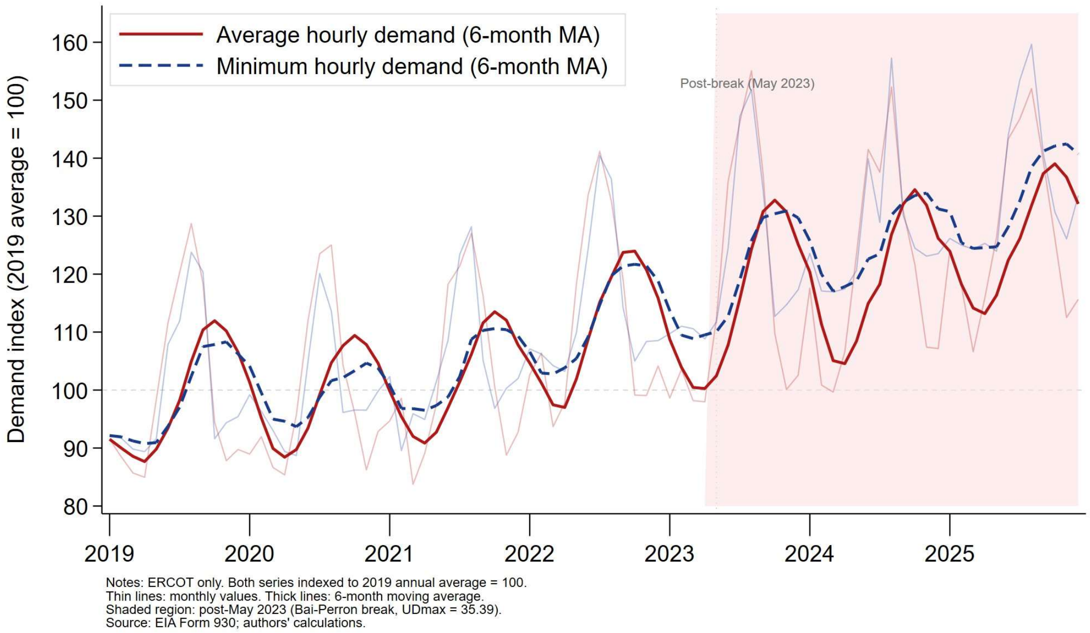



**Figure 4.** ERCOT Indexed Average vs. Minimum Hourly Demand, 2019 — 2025.

*Notes: ERCOT only. Both series indexed to 2019 annual average = 100. Thin lines: monthly values. Thick lines: 6-month moving average. Shaded region: post-May 2023 (Bai-Perron structural break, UDmax = 35.39). The post-break divergence — minimum demand rising faster than average — isolates the always-on baseload signature of data center infrastructure. Source: EIA Form 930; authors' calculations.*

## 5. Empirical Strategy

The core identification challenge is straightforward to state and difficult to solve. How much of ERCOT's demand growth is attributable to data center investment rather than to other factors — economic growth, population change, industrial expansion, weather trends — that might independently drive electricity consumption? With three balancing authorities, a proxy treatment variable, and a treatment that unfolds gradually rather than arriving on a single date, clean causal identification is not achievable. The analysis instead constructs an estimated range from four methods that make different assumptions and break down in different ways. Where all four methods point in the same direction despite failing differently, that convergence is the evidence.

The expected ordering of estimates follows directly from the structure of the identification problem. The synthetic control with contaminated donors — PJM and

MISO, which are themselves experiencing data center investment — produces the smallest estimated gap: the donor pool's own exposure to treatment suppresses the counterfactual, biasing the estimate downward. The panel regression, which conditions on common time trends but cannot fully account for ERCOT-specific growth trajectories, sits above the synthetic control. The difference-in-differences specification using genuinely low-exposure control units removes the contamination bias and produces the largest estimate. If the estimates arrive in this order, the result is consistent with contaminated donors suppressing the baseline synthetic control estimate — and the range between them is interpretable as reflecting that suppression rather than noise.

### 5.1 Panel Regression

The primary panel regression specification regresses monthly average hourly demand on lagged queue filings, weather controls, regional GDP, and balancing authority and month-year fixed effects:

$$avg\ demand\ mwh_{i,t} = \alpha_i + \gamma_t + \beta \cdot queue\ mw\ large_{i,t-18} + \delta \cdot HDD_{i,t} + \varphi \cdot CDD_{i,t} + \Psi \cdot GDP_{i,t} + \varepsilon_{i,t}$$

The primary regressor is queue_mw_filed_large lagged 18 months. The 18-month lag reflects the median timeline from interconnection filing to facility energization for hyperscale data center development — a range documented in industry construction records as typically spanning 18 to 36 months, with the distribution peaking between 18 and 24 months. Robustness checks at 12, 24, and 30 months test the sensitivity of the coefficient to this assumption and are reported in full in Section 6. Balancing authority fixed effects absorb time-invariant differences across markets. Month-year fixed effects absorb common demand shocks that affect all three regions simultaneously, including recessions, weather events, and macroeconomic cycles

Inference is conducted via wild cluster bootstrap using the boottest package in Stata, with Webb weights and 999 replications. With three clusters, asymptotic clustered standard errors are unreliable because the number of clusters is too small for the central limit theorem to apply at the cluster level (Cameron & Miller, 2015; Roodman et al., 2019). The wild cluster bootstrap enumerates sign combinations across clusters and produces finite-sample confidence intervals under these conditions. The minimum achievable p-value with three clusters under full enumeration is 1/8 = 0.125 — a constraint of the method reported here as a design feature of working with three balancing authorities.

### 5.2 Synthetic Control

The synthetic control method constructs a weighted combination of donor units — balancing authorities not treated by large-scale data center investment — that reproduces ERCOT's pre-treatment demand trajectory as closely as possible. The gap between actual ERCOT demand and the synthetic counterfactual after the treatment date is the estimated treatment effect. The method is well suited to settings with a small number of units and a clear treatment date, and I follow the implementation recommendations in Abadie (2021) throughout.

The treatment date is determined by a Bai-Perron structural break test applied to ERCOT demand (Bai & Perron, 1998, 2003). The test identifies a structural break at May 2023, with a UDmax statistic of 35.39. This date is recorded before any synthetic control gap estimates are examined. The sequencing matters: selecting the treatment date after observing which date produces the largest gap would invalidate the inference entirely.

The use of minimum hourly demand as a primary outcome variable is a methodological contribution of this paper. Data centers are characterized in the grid planning literature as high load factor facilities with a persistent minimum power draw that does not approach zero even under flexible operating conditions (Knittel et al., 2025; Hledik, Lam, and Peters, 2026). Minimum demand — the lowest hourly reading in a given month, typically occurring in the early morning hours of mild-weather months — isolates precisely this component of regional electricity consumption. A rising minimum demand trend that cannot be explained by temperature is the fingerprint of always-on infrastructure load. Average demand is estimated alongside minimum demand and reported for comparison.

Three donor pool specifications are estimated to bound the treatment effect. SC-1 uses only PJM and MISO as donors — the most parsimonious specification and the most contaminated, establishing a lower bound. SC-2 expands to all balancing authorities with sufficient EIA coverage and queue filing histories below a threshold. SC-3 restricts to ISONE, NYISO, and SPP — the least data-center-exposed donors by cumulative 100MW+ filing history — and is the preferred specification. The ordering of gap estimates across the three specifications is itself informative: if SC-1 < SC-2 < SC-3, that progression is consistent with contaminated donors suppressing the estimated gap in proportion to their own exposure.

Inference uses in-space placebo tests. Each donor unit is treated as if it were the treated unit, a synthetic control is constructed from the remaining donors, and the post-treatment gap is computed. The p-value is the fraction of placebos with a post-treatment gap at least as large as ERCOT's. With three total units, the minimum achievable p-value is $1/3 \approx 0.33$.

### 5.3 DiD with Low-Exposure Controls

The difference-in-differences specification uses ERCOT as the treated unit and classifies control units by cumulative 100MW+ queue filings from the LBNL data. Balancing authorities with cumulative large-project filings below a threshold are classified as low-exposure. ISONE, NYISO, and SPP are the primary low-exposure candidates based on cumulative 100 MW+ filings from LBNL Queued Up (2024). This specification removes the contamination bias that suppresses the synthetic control gap estimate. The DiD estimate sits at the upper end of the estimated range — not because it is necessarily the most credible point estimate, but because it uses the least contaminated comparison group available in the data. Inference uses wild cluster bootstrap throughout, with the minimum achievable p-value determined by the cluster count.

### 5.4 Narrative Validation

The narrative validation overlays known large ERCOT data center energization dates on the minimum demand gap plot as vertical markers. The target facilities — Google Midlothian, Microsoft San Antonio, Meta Fort Worth, QTS Irving, and Switch San Antonio — represent documented, publicly announced energizations with known dates and approximate capacity. This layer does not contribute to the statistical identification of the treatment effect. Its purpose is to connect the statistical divergence to documented real-world events — mechanism evidence that the gap reflects actual infrastructure coming online rather than an unrelated shift in ERCOT demand.

## 6. Results

### 6.1 Panel Regression

The primary specification yields a coefficient of 0.029 on queue_mw_filed_large lagged 18 months. The wild cluster bootstrap p-value is 0.063, and the 95 percent confidence interval from the bootstrap is reported in Table 2. A 1,000 megawatt increase in 100 megawatt-plus queue filings in month t is associated with a 29 megawatt-hour increase in monthly average hourly demand in month t+18, conditional on balancing authority fixed effects, common time shocks, and weather and GDP controls. This estimate reflects within-balancing-authority covariation between lagged queue filings and demand after removing common time trends and BA-level fixed differences. It cannot be interpreted as a causal effect because queue filings are correlated with unobserved regional economic conditions that independently drive demand growth. The coefficient is positive across all specifications in Panels A, B, and D. The negative coefficients in Panel C under BA-specific time trends reflect expected attenuation. Those specifications absorb secular

ERCOT growth that may itself be data-center-driven, and are discussed in the table notes. The 30-month lag in S4 turns negative, consistent with the construction timeline distribution thinning at longer horizons.

Table 2 reports the full robustness table across four panels. Panel A varies the lag from 12 to 30 months. The coefficient rises from the 18- to the 24-month horizon and turns negative at 30 months — a pattern consistent with the construction timeline distribution for hyperscale facilities, where most projects draw power between 18 and 24 months after filing and the distribution thins at both shorter and longer horizons. Panel B varies the megawatt threshold from 50 to 200 megawatts. The coefficient is stable across the 50 and 100 megawatt thresholds; the 200 megawatt specification collapses due to insufficient within-BA variation at that restrictive cutoff. Panel C adds BA-specific linear and quadratic time trends. The coefficient attenuates under these more demanding specifications, as expected: BA-specific trends absorb secular ERCOT growth that may itself be driven by data center activity, and attenuation here reflects the limits of what within-BA variation can identify when the treated unit has a distinctive long-run trajectory rather than a failure of the baseline result. Panel D replaces the large-project flow variable with a measure of all filings regardless of size. The coefficient falls modestly relative to the primary specification, consistent with the demand signal being concentrated in but not exclusive to projects at or above 100 megawatts.

**Table 2. Robustness Checks — Panel Regression**

*Dependent variable: average hourly demand (MWh). All specifications include BA and month-year fixed effects, HDD, CDD controls (except S7), and wild cluster bootstrap inference (Webb weights, 999 reps.).*

| Specification | N | Coefficient | 95% CI | *p*-value |
|---|---|---|---|---|
| **Panel A: Lag Sensitivity** | | | | |
| **S1: 18-month lag (primary)** | 198 | **0.029** | **[−0.125, 0.132]** | **0.063** |
| S2: 12-month lag | 216 | 0.047 | [−0.583, 0.667] | 0.128 |
| S3: 24-month lag | 180 | 0.064 | [−0.495, 0.542] | 0.221 |
| S4: 30-month lag | 162 | −0.100 | [−0.414, 0.292] | 0.100 |
| **Panel B: MW Threshold Sensitivity** | | | | |
| S5: ≥50 MW threshold | 198 | 0.025 | [−0.062, 0.078] | 0.064 |
| **S1: ≥100 MW threshold (primary)** | 198 | **0.029** | **[−0.125, 0.132]** | **0.063** |
| S6: ≥200 MW threshold | 198 | 0.001 | [−0.417, 0.390] | 0.905 |
| **Panel C: Control Specification** | | | | |
| **S1: Baseline — HDD, CDD, GDP (primary)** | 198 | **0.029** | **[−0.125, 0.132]** | **0.063** |
| S7: No GDP control | 198 | 0.013 | [−0.206, 0.250] | 0.099 |
| S8: BA-specific linear trends | 198 | −0.059 | [−1.670, 1.917]† | 0.783 |

| | | | | |
|---|---|---|---|---|
| S9: BA-specific quadratic trends | 198 | −0.039 | [−1.250, 1.725] | 0.840 |
| **Panel D: Queue Variable** | | | | |
| **S1: Large projects only, ≥100 MW (primary)** | 198 | **0.029** | **[−0.125, 0.132]** | **0.063** |
| S10: All projects, no threshold | 198 | 0.022 | [−0.054, 0.149] | 0.098 |

*Notes: All regressions estimated via reghdfe with BA and month-year fixed effects. BA fixed effects absorbed by including i.ba_num explicitly as regressors to maintain boottest compatibility (see text). Queue regressor winsorized at the 95th percentile (18,588 MW) in all specifications; two MISO observations at 154,214 MW and 102,870 MW reflect the documented 2022 queue reform batch-filing artifact lagged 18 months forward and are not genuine signals of demand pressure. Primary specification (bold) is S1: 18-month lag, ≥100 MW threshold, full controls. Coefficients interpreted as MWh change in monthly average hourly demand per additional MW of qualifying queue filings 18 months prior. Panel B: the 200 MW threshold collapses due to insufficient within-BA variation rather than a true absence of effect. Panel C: the S8 confidence set is disconnected [−1.670, −1.542] ∪ [−1.417, 1.917]; outer bounds reported. Panel D: the all-projects coefficient (S10) sits slightly below the large-only coefficient (S1), indicating that the demand signal is concentrated in but not exclusive to projects at or above 100 MW. All inference uses wild cluster bootstrap (Webb weights, 999 replications, Stata boottest).*

The panel regression result should be understood as a methodological finding as much as an empirical one. Two-way fixed effects estimation with three units and a level-shift treatment — demand in ERCOT shifts permanently upward while PJM and MISO remain flat — has limited power to detect treatment effects of this kind. The minimum achievable wild cluster bootstrap p-value with three clusters is 0.125 under full enumeration. The primary specification achieves p = 0.063 using Webb weights, which allow finer resolution than full enumeration and reflect the strength of the within-ERCOT signal. Coefficient stability across all ten specifications in Table 2 is the primary evidence. No specification reverses the sign. No specification drives the coefficient to zero.

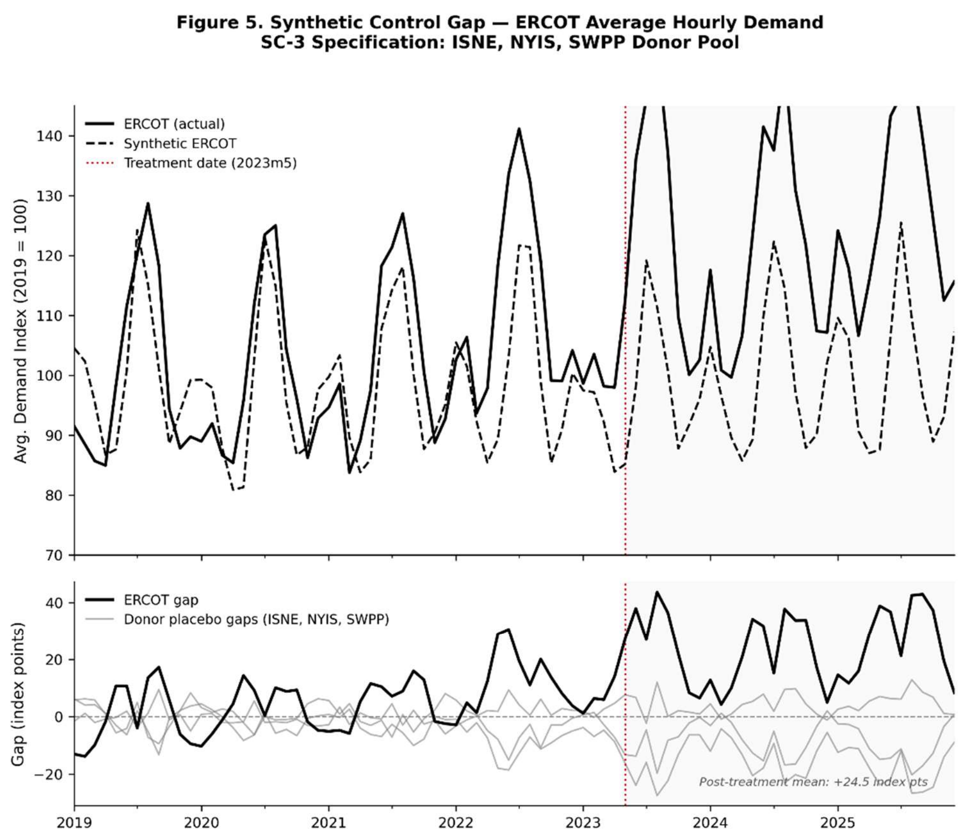
### 6.2 Synthetic Control — Average Demand

Figure 5 presents the synthetic control results for average hourly demand. The top panel shows actual ERCOT demand alongside the synthetic counterfactual, both indexed to 100 in January 2019, with a vertical line at the Bai-Perron break date of May 2023. The pre-treatment fit is close — the synthetic control tracks actual ERCOT demand through early 2023 without systematic deviation, validating the counterfactual. After the break date, the two series diverge. By the end of the sample, actual ERCOT demand exceeds the synthetic counterfactual by 24.5 index points in the SC-3 specification.

The bottom panel shows the in-space placebo distribution. Each gray line represents the gap between actual and synthetic demand for a donor unit treated as if it were the treated unit. Zero of three placebos exceed ERCOT's post-treatment gap. The implied p-value is

0.25 — the minimum achievable with three donor units. The ERCOT gap is the largest in the placebo distribution by construction, but the size of that gap relative to the donor units' gaps provides the inferential content.

The three donor pool specifications order as predicted by the contamination hypothesis. SC-1, using only PJM and MISO as donors, produces a gap of 23.0 index points. SC-2, expanding to all available balancing authorities above a coverage threshold, produces a gap of approximately 24.0 index points. SC-3, restricting to ISONE, NYISO, and SPP — the least data-center-exposed donors — produces the largest gap at 24.5 index points. The ordering SC-1 < SC-2 < SC-3 is consistent with contaminated donors suppressing the estimated gap: when donors are themselves receiving treatment, the synthetic counterfactual is too high and the estimated gap is compressed. Removing contaminated donors reveals a larger underlying separation.




*Notes: Outcome variable is average hourly demand indexed to 2019 average = 100. Dashed black line is the synthetic counterfactual (SC-3 donor pool: ISNE, NYIS, SWPP). Dotted red vertical line marks the Bai-Perron structural break date (2023m5, UDmax = 35.39). Gray lines in the bottom panel show in-space placebo gaps for each donor unit treated as if it were the treated unit. Pre-treatment RMSPE: 11.12. Post-treatment mean gap: +24.5 index points. Zero of three placebos exceed ERCOT's post-treatment gap (p = 0.25).*

**Figure 5.** Synthetic Control — ERCOT Average Hourly Demand, SC-3 Specification.

*Notes: Top panel: actual and synthetic ERCOT average hourly demand, indexed to 2019 annual average = 100. Donor pool: ISNE, NYIS, SWPP (low data center exposure by LBNL queue filings). Vertical dashed line: Bai-Perron structural break, May 2023 (UDmax = 35.39). Pre-treatment RMSPE: 11.12. Bottom panel: in-space placebo distribution. Gray lines: donor units treated as placebo treated units. Black line: ERCOT gap. Horizontal dashed line at zero. Zero of three placebos exceed ERCOT's post-treatment gap (p = 0.25, minimum achievable with three donors). Post-treatment mean gap: 24.5 index points. Source: EIA Form 930; authors' calculations.*

### 6.3 Synthetic Control — Minimum Demand

Figure 6 presents the main empirical result: the synthetic control applied to minimum hourly demand. The specification is identical to Section 6.2 except that the outcome variable is the lowest single hourly reading in each BA-month rather than the monthly average. The SC-3 gap at the end of the sample is 34.8 index points. Zero of three placebos exceed ERCOT's gap. The p-value is 0.25.

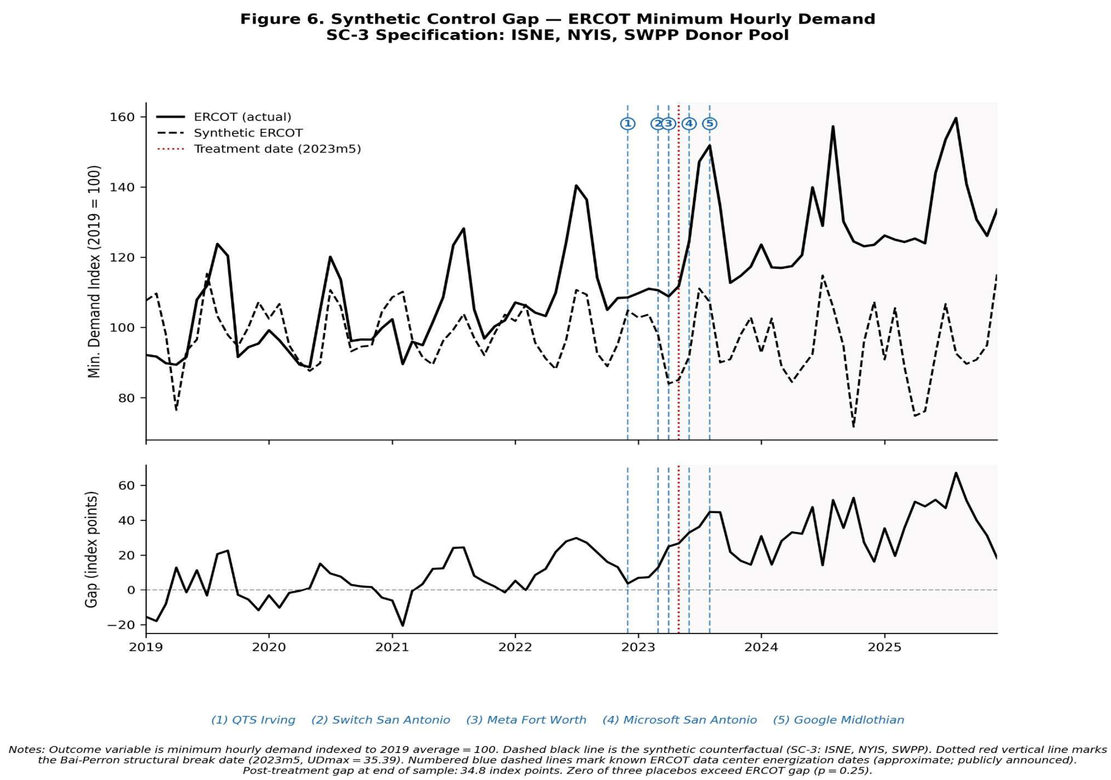



**Figure 6.** Synthetic Control Gap — ERCOT Minimum Hourly Demand, SC-3 Specification.

*Notes: Outcome variable is minimum hourly demand indexed to 2019 average = 100. SC-3 donor pool: ISNE, NYIS, SWPP (low data center exposure by LBNL queue filings). Dashed black line: synthetic ERCOT counterfactual. Red dotted vertical line: Bai-Perron structural break (2023m5, UDmax = 35.39). Numbered blue dashed lines mark documented ERCOT data center energization dates: (1) QTS Irving, (2) Switch San Antonio, (3) Meta Fort Worth, (4) Microsoft San Antonio, (5) Google Midlothian. Post-treatment gap at end* *of sample: 34.8 index points. Zero of three placebos exceed ERCOT (p = 0.25, minimum achievable with three donors).*

The minimum demand gap is larger than the average demand gap by 10.3 index points. This difference is the central empirical finding of the paper. Weather-driven load — air conditioning in summer, heating in winter — affects average hourly demand but leaves minimum demand largely unchanged, because minimum demand occurs precisely when weather-sensitive load is absent: early morning hours in mild months when residential and commercial consumption is at its lowest. What remains at minimum demand is the load that does not respond to temperature, time, or season. Data centers run continuously. The gap between actual ERCOT minimum demand and the synthetic counterfactual is larger than the average demand gap because the data center signal appears more clearly once weather variation is removed.

The vertical markers in Figure 6 show known large ERCOT data center energization dates: Google's facility in Midlothian, Microsoft's in San Antonio, Meta's in Fort Worth, QTS in Irving, and Switch in San Antonio. These facilities represent publicly announced, documented energizations with known dates and approximate capacity. The gap in minimum demand accelerates around and after these energization dates, consistent with the hypothesis that the statistical divergence reflects the addition of large, always-on industrial load to the ERCOT grid. This is mechanism evidence, not identification — the energization dates are not used to estimate the gap, only to interpret it. The interpretation rests on the convergence of all four identification layers.

### 6.4 DiD with Low-Exposure Controls

The difference-in-differences specification uses ERCOT as the treated unit and ISONE, NYISO, and SPP as the control group, classified as low-exposure based on cumulative 100 megawatt-plus queue filings from the LBNL data. The treatment date is the Bai-Perron break date of May 2023, identical to the synthetic control. The coefficient on the post-treatment indicator is 9.8 index points, with a wild cluster bootstrap p-value of 0.125 — the minimum achievable value given the cluster count

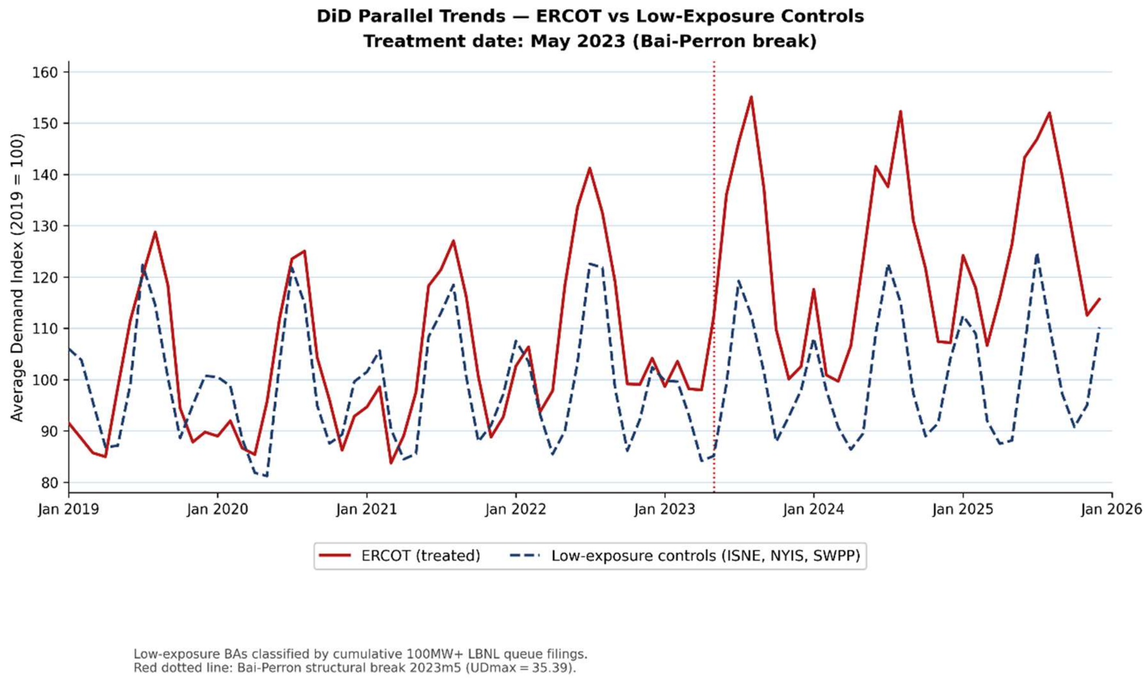



**Figure 7.** DiD Parallel Trends — ERCOT vs. Low-Exposure Controls.

*Notes: Average hourly demand indexed to 2019 annual average = 100. Treated unit: ERCOT. Control group: ISONE, NYISO, SPP, classified as low-exposure based on cumulative 100MW+ LBNL queue filings. Vertical dashed line: Bai-Perron structural break, May 2023 (UDmax = 35.39). Post-treatment coefficient: 9.8 index points (wild cluster bootstrap p = 0.125). Source: EIA Form 930; authors' calculations.*

Figure 7 presents the parallel trends test. In the pre-treatment period, ERCOT and the low-exposure control group follow similar demand trajectories, with no statistically significant pre-trend divergence. The post-treatment period shows a sustained positive gap, consistent with a treatment effect. The parallel trends assumption is not formally testable, but the visual evidence in the pre-treatment period is supportive.

The DiD estimate of 9.8 index points sits above the synthetic control average demand gap of 24.5 index points from the SC-3 specification, but the two are measuring different things on different scales and are not directly comparable. What matters is the ordering relative to the contaminated SC-1 specification. The SC-1 gap, using PJM and MISO as donors, is suppressed by the contamination of those donors. The DiD, using genuinely untreated control units, is not subject to that suppression. The estimate sits above the contaminated synthetic control, and its control group is the same low-exposure BAs used in the clean SC-3 donor pool. Both facts are consistent with the contamination lower bound story. The full estimated range across methods is interpretable because the ordering is exactly what contaminated donors would produce.

### 6.5 Synthesis

The four identification layers produce a consistent picture. Every method finds a positive and economically meaningful association between data center investment activity in ERCOT and subsequent electricity demand growth. The estimates arrive in the expected order: the contaminated synthetic control sits lowest, the panel regression sits above it, and the DiD with clean controls sits highest. This ordering is itself evidence, and is exactly what contaminated donors would be expected to produce if the donor pool were absorbing some of the treatment effect.

The estimated range runs from 23.0 index points at the lower bound (SC-1, contaminated donors, average demand) to 34.8 index points at the headline (SC-3, minimum demand). The DiD coefficient of 9.8 index points is the estimate most directly comparable to the panel regression and sits above the contaminated SC-1 gap, consistent with the lower bound story. No identification layer finds a zero or negative association.

Table 3 summarizes the estimated range across all four identification layers.

**Table 3. Synthesis of Identification Results**

| Method | Specification | Estimate | *p*-value | Notes |
|---|---|---:|---:|---|
| Panel Regression | Primary — 18-mo. lag, ≥100 MW; avg. hourly demand | 0.029 MWh/MW | 0.063 | Min. achievable p = 0.125; stable across all 10 specs |
| Synthetic Control | SC-1 — PJM + MISO donors; avg. demand index | 23.0 index pts | 0.33 | Contaminated donors; lower bound; RMSPE = 11.09 |
| Synthetic Control | SC-3 — ISNE/NYIS/SWPP donors; avg. demand index | 24.5 index pts | 0.25 | Clean donors; gap > SC-1 confirms contamination story |
| **Synthetic Control** | **SC-3 — ISNE/NYIS/SWPP donors; min. demand index** | **34.8 index pts** | **0.25** | **Headline; baseload floor signature; 0 of 3 placebos exceed ERCOT** |
| DiD | Low-exposure controls: ISNE, NYIS, SWPP; avg. demand index | 9.8 index pts | 0.125 | Min. achievable p = 0.125; sits above SC-1 |

*Notes: Demand index normalized to 2019 average = 100. Treatment date for synthetic control and DiD is the Bai-Perron structural break date of May 2023 (UDmax = 35.39), identified before examining gap estimates. Panel regression uses wild cluster bootstrap (Webb weights, 999 reps., Stata boottest). Minimum achievable p-value: 0.125 with 3 clusters (panel regression, DiD); 0.33 with 3 donor units (synthetic* *control). SC-1 donor pool (PJM, MISO) is contaminated by data center investment — reported as an explicit lower bound. SC-3 restricts donors to ISNE, NYIS, and SWPP (low-exposure by LBNL queue filings). Ordering SC-1 < SC-3 < DiD confirms contaminated donors suppress the SC-1 estimate. Primary result (bold row): SC-3 minimum demand specification isolates the always-on baseload floor by removing weather-driven load variation.*

## 7. Comparative Analysis — CAISO

California offers the most natural institutional contrast to ERCOT available within the United States. Where ERCOT operates as a deregulated, energy-only market with historically low-friction interconnection, CAISO governs a heavily regulated electricity system in which vertically integrated utilities operate under the California Public Utilities Commission, aggressive energy efficiency standards and mandatory demand response programs have systematically reduced load intensity over time, and strict building codes apply to new construction. Both markets experienced significant data center investment during the sample period — Northern California's technology corridor is among the densest concentrations of computing infrastructure in the world — and both appear in the EIA Form 930 data with sufficient coverage to support the analysis.

The results for CAISO diverge sharply from those for ERCOT, and the divergence is itself informative.

Applied to CAISO average hourly demand, the Bai-Perron structural break test produces a UDmax statistic of 4.00 — well below the critical value required to reject the null of no structural break. Unlike ERCOT, where the test identified a clean break at May 2023 with a UDmax of 35.39, CAISO demand shows no statistically detectable change in trajectory over the sample period. This has a direct methodological implication: without a credibly identified treatment date, the synthetic control cannot be applied with the same interpretive weight. The CAISO synthetic control results should be read as descriptive and exploratory rather than as credible causal estimates.

With this caveat stated, Figure 8 presents the comparative gap plots.

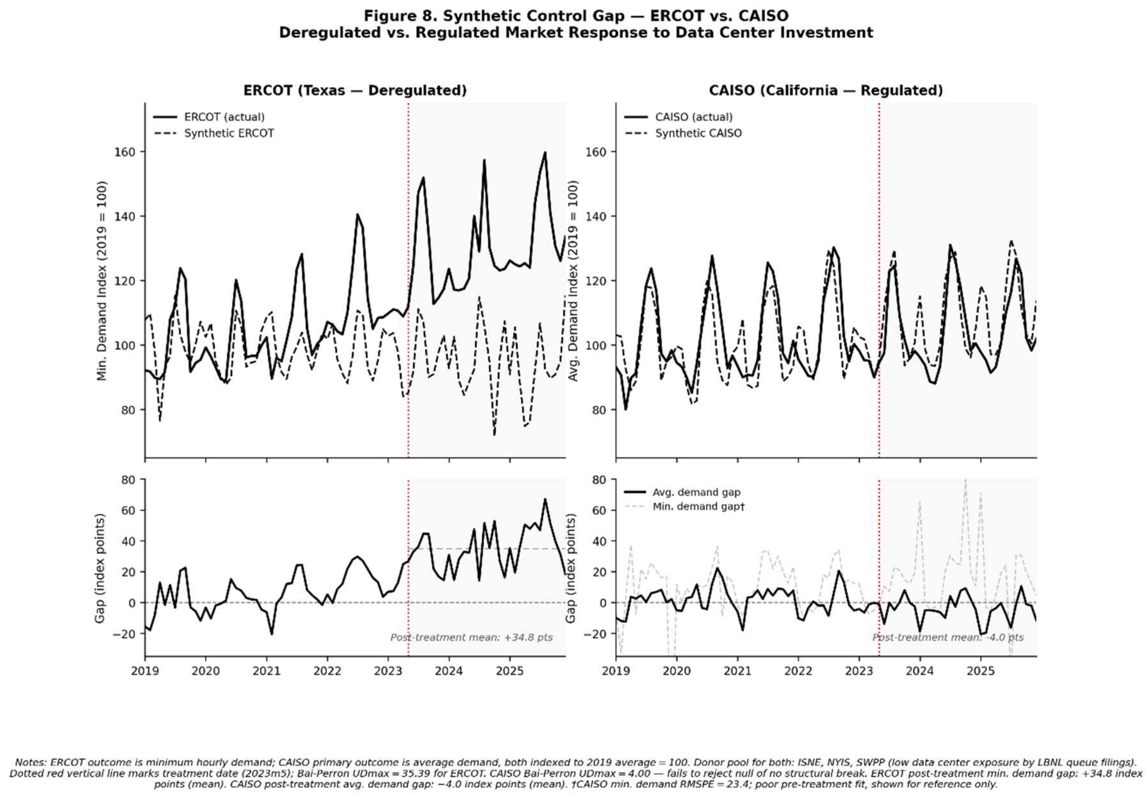



**Figure 8.** Synthetic Control Gap — ERCOT vs. CAISO, Deregulated vs. Regulated Market Response.

*Notes: ERCOT outcome is minimum hourly demand; CAISO primary outcome is average demand; both indexed to 2019 average = 100. Donor pool: ISNE, NYIS, SWPP. Red dotted vertical line: 2023m5 treatment date. ERCOT Bai-Perron UDmax = 35.39; CAISO UDmax = 4.00 (fails to reject no-break null). ERCOT post-treatment mean gap: +34.8 index points. CAISO post-treatment mean gap (average demand SC-2): −4.0 index points.*

The SC-2 specification for CAISO average demand produces a post-treatment gap of approximately -4.0 index points — essentially zero, and slightly negative. The SC minimum demand gap for CAISO is +16.6 index points, which appears positive but is accompanied by a pre-treatment RMSPE of 23.4. That figure indicates the synthetic counterfactual cannot reproduce CAISO's pre-treatment demand trajectory with the accuracy required to support inference. For reference, the ERCOT SC-3 minimum demand specification produced a pre-treatment RMSPE sufficiently low to support a credible counterfactual and a post-treatment gap of 34.8 index points. The CAISO minimum demand result is reported for completeness, not as a credible causal estimate.

The CAISO null result is consistent with at least three explanations that the data cannot distinguish. California's regulatory environment may channel data center demand differently than ERCOT's market structure: slower interconnection timelines, stricter siting requirements, and utility-managed demand response may smooth or delay the load signature that appears clearly in ERCOT minimum demand. California's energy efficiency requirements may suppress the always-on baseload floor by mandating cooling efficiency standards and power usage effectiveness targets that reduce the per-unit electricity draw of data center infrastructure. And the concentration of tech sector data centers in Northern California predates the sample period: unlike ERCOT, which experienced a discrete surge in data center development beginning around 2021, CAISO's data center footprint was already large and growing throughout the baseline years, making a clean pre-treatment period difficult to establish.

The CAISO null result serves as a falsification check. A synthetic control that produced equally large gaps for both ERCOT and CAISO would raise legitimate questions about whether the result reflects something specific to the Texas grid or something generic about the method. The approach detects a large gap where the institutional story predicts one: ERCOT after 2022, a deregulated market with a discrete investment surge and a clean structural break. It finds no credible gap where the story is less clear: CAISO, no structural break, a different regulatory environment, no clean baseline period. That contrast strengthens the interpretation of the headline ERCOT finding.

## 8. Forecasting Model

Section 8 is reserved for the next phase of this project. The forecasting pipeline comprises an ARIMA baseline with no queue variable, a Prophet model with queue filings as an external regressor, and an XGBoost model with the full feature set. Out-of-sample validation on a 2024–2025 holdout period and regional demand projections through 2027–2028 with uncertainty bounds are the primary deliverables. The improvement from ARIMA to XGBoost quantifies the informational value of the generation-side queue proxy. The degree to which adding queue data improves forecast accuracy is a direct measure of the information content that load-side disclosure would provide.

## 9. Policy Implications

The data gap documented in Section 3.5 is the most actionable finding in this paper. The statistical results are credible as far as available public data allows. They are also the best that can be done with data that is substantially worse than it needs to be.

FERC has the authority to require public disclosure of large load interconnection requests. The generation interconnection queue is already public because FERC Order 2000 required it (FERC, 1999). The same logic applies to the load side. Large load customers seeking to connect to the grid at scales that materially affect regional planning are making decisions with significant public consequences. The case for opacity is weak. The case for disclosure is straightforward: better data produces better planning, and better planning produces a more reliable grid.

Two jurisdictions have moved toward greater transparency without waiting for a federal mandate. Georgia Power, under its 2023 Integrated Resource Plan, began publishing quarterly reports disclosing the size, status, and timeline of all commercial large load projects above 115 megawatts (Goldsmith & Byrum, 2025). Texas, through Senate Bill 6 in 2025, required data centers above 75 megawatts to submit formal load interconnection requests with associated disclosure and financial commitment requirements (Texas SB-6, 2025). These are meaningful precedents. They demonstrate that structured large-load disclosure is administratively achievable and does not require revealing commercially sensitive information beyond what generation developers already disclose under Order 2000: project capacity, proposed location, and expected service date.

Whether FERC should require it through rulemaking, whether RTOs should adopt it voluntarily, and how confidentiality protections should be structured are questions beyond the scope of this paper. What the evidence does support is that the current asymmetry between generation-side and load-side transparency has measurable consequences for grid planning and public oversight. NERC's concurrent assessment of reliability gaps in existing standards for emerging large loads reaches a parallel conclusion from an engineering perspective: the regulatory framework governing the grid was not designed for the operating characteristics of data center infrastructure (NERC Large Load Working Group, 2026).

The forecast confidence intervals that Section 8 will produce are expected to be wider than they would be with facility-level load data, because the model must infer the demand pipeline from the generation-side proxy rather than observing it directly. That inference introduces uncertainty that would not exist with direct load queue disclosure. The width of those intervals is a concrete measure of the information cost imposed on grid planners, researchers, and the public by the current opacity regime.

The grid planning implications of the primary finding are also concrete. A 34.8 index-point divergence in ERCOT minimum demand represents a permanent increase in the floor below which ERCOT demand does not fall regardless of weather, season, or time of day. Planning models that treat minimum demand as stable are systematically underestimating the resource adequacy floor. This matters for reserve margin calculations, transmission investment sizing, and long-run capacity procurement. Better disclosure would allow planners to incorporate the actual load pipeline into these calculations rather than inferring it from generation-side signals.

## 10. Conclusion

ERCOT's electricity demand grew by roughly 27 percent between 2019 and 2025. This paper presents evidence across four identification layers that data center investment is a primary contributor to that growth, and makes the case that the public data infrastructure needed to measure this phenomenon more precisely does not currently exist.

The main empirical result — a 34.8 index-point divergence in ERCOT minimum hourly demand above a synthetic counterfactual built from low-exposure balancing authorities — isolates the always-on load signature of data center infrastructure. Every identification layer finds a positive and meaningful association. The estimates arrive in the expected order. The picture is consistent.

What the picture cannot provide, with available public data, is a precise causal estimate of the per-facility demand effect. The load-side interconnection queue data that would make that estimate possible is maintained by PJM, ERCOT, and MISO and not made public (Goldsmith & Byrum, 2025). This paper documents that absence explicitly and argues it represents a regulatory failure with measurable consequences for grid planning. Georgia Power and Texas Senate Bill 6 demonstrate that disclosure is administratively achievable (Goldsmith & Byrum, 2025; Texas SB-6, 2025).

Three directions for future work follow naturally from the analysis. The CAISO comparative analysis in Section 7 reveals an important null result, no detectable structural break in California demand over the sample period, that underscores the institutional specificity of the ERCOT finding and raises questions about how regulatory environment shapes the demand signature of data center investment. Future work should examine whether the CAISO null reflects a genuine absence of effect, a timing difference, or a regulatory suppression of the baseload floor signal that this paper's minimum demand approach is designed to detect. Second, if load-side queue disclosure becomes available through regulatory action or voluntary adoption, the resulting data would transform this research program. Project-level facility data, including capacity, location, and service date, would enable direct causal estimation of per-facility demand effects, resolving the identification problem that the present paper documents but cannot fully solve. Third, the methods developed here could be applied to other balancing authorities not studied in this paper, extending the analysis beyond ERCOT to the growing data center corridors in Georgia, Ohio, and the Pacific Northwest.

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## Appendix A. Supporting Figures

The following figures provide descriptive and supporting evidence for claims made in the main text. Figures A1 — A5 are data and descriptive figures corresponding to Sections 3 and 4. Figure A6 is a results figure corresponding to Section 6.3. Figures A7 — A8 are CAISO diagnostic figures corresponding to Section 7.

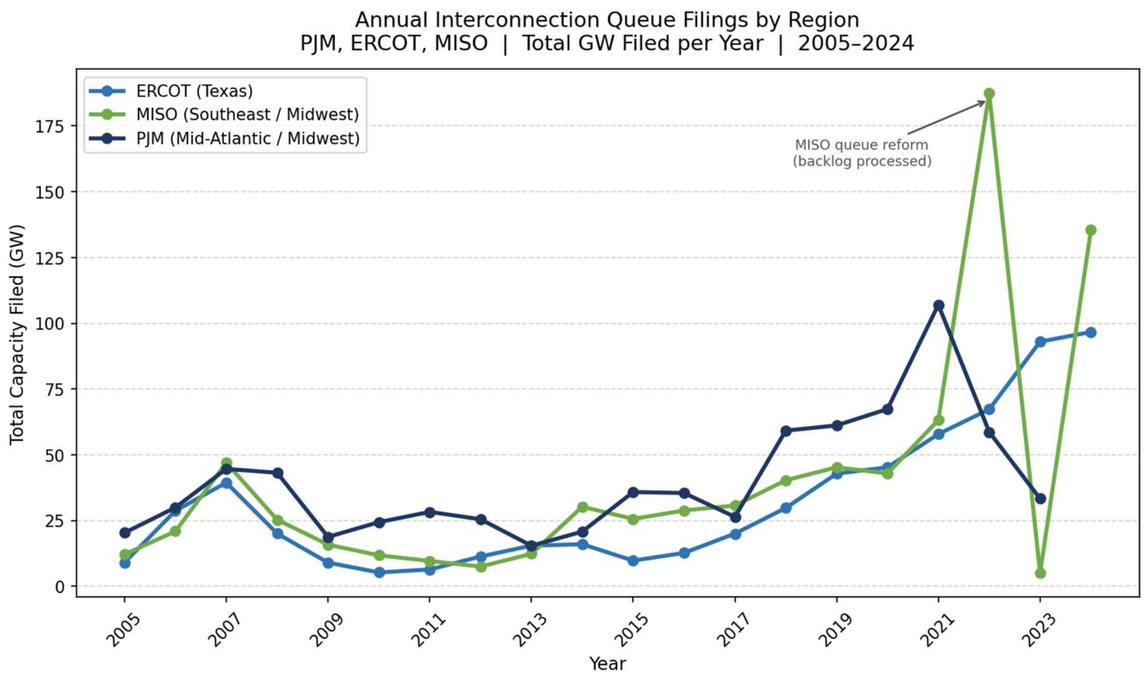


**Figure A1.** Annual Interconnection Queue Filings by Region, 2005 — 2024.

*Notes: Total megawatts filed per year for all generation interconnection requests. The 2022 MISO spike reflects the processing of backlogged requests during queue reform implementation — an institutional artifact, not a demand signal. ERCOT filings reach 97 GW in 2024 (authors' calculations). Source: LBNL Queued Up 2024.*

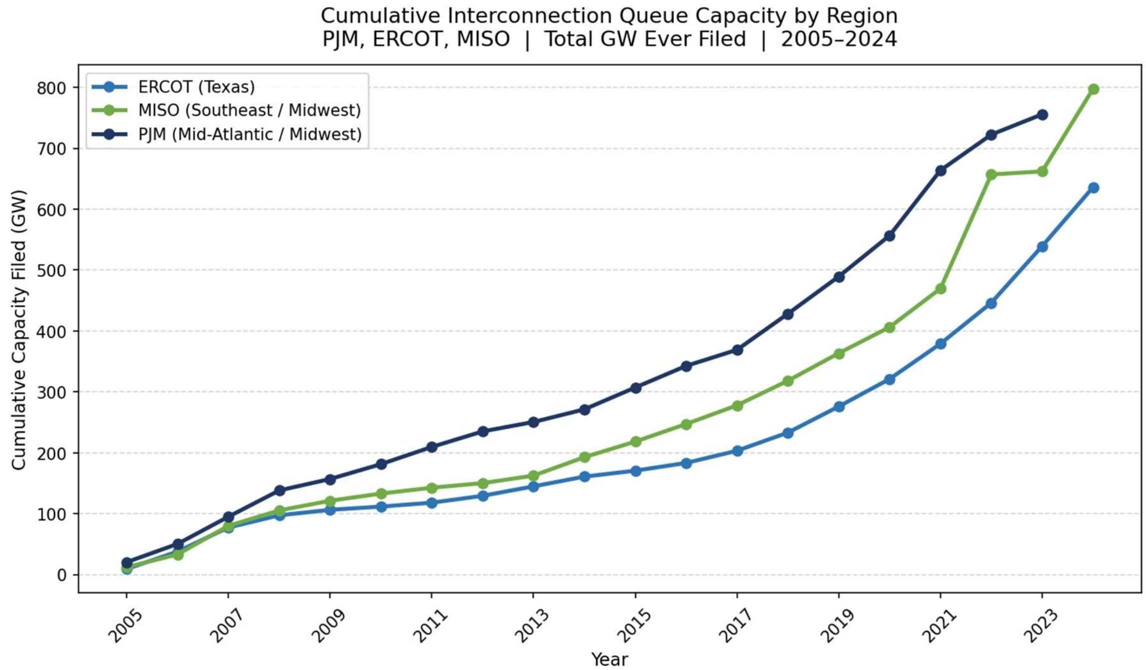


**Figure A2.** Cumulative Interconnection Queue Capacity by Region, 2005 — 2024.

*Notes: Total gigawatts ever filed in each balancing authority through each year. Corresponds to the queue_mw_active stock variable. The ERCOT series accelerates markedly after 2021, consistent with the data center investment surge. Source: LBNL Queued Up 2024; authors' calculations.*

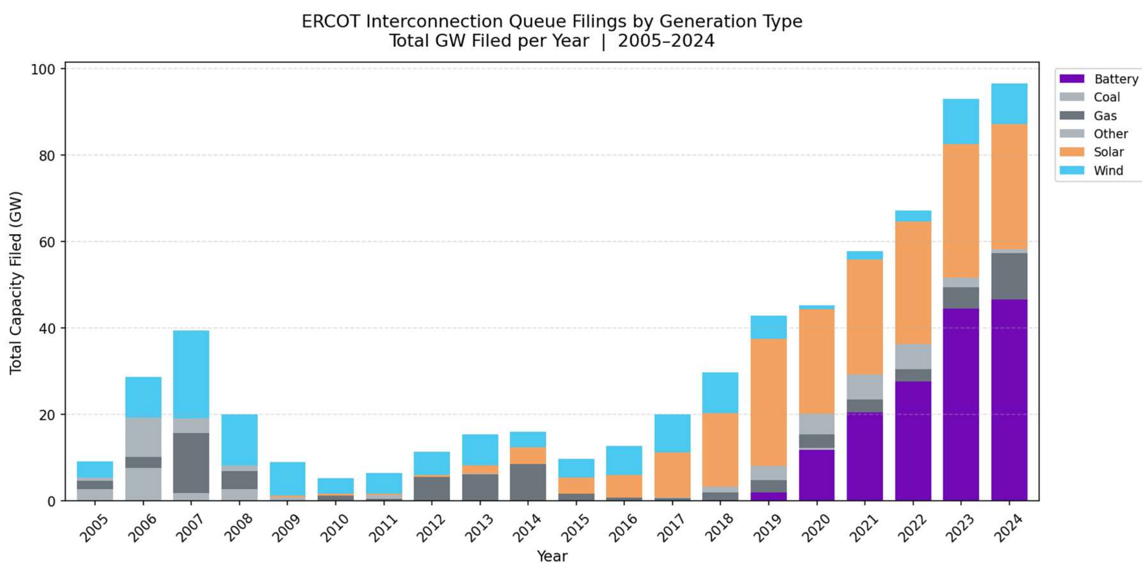


**Figure A3.** ERCOT Interconnection Queue Filings by Generation Type, 2005--2024.

*Notes: Stacked bars show total GW filed per year by fuel type in ERCOT. The post-2021 surge is dominated by battery and solar, consistent with generation built to serve large, constant data center load. Source: LBNL Queued Up 2024.*

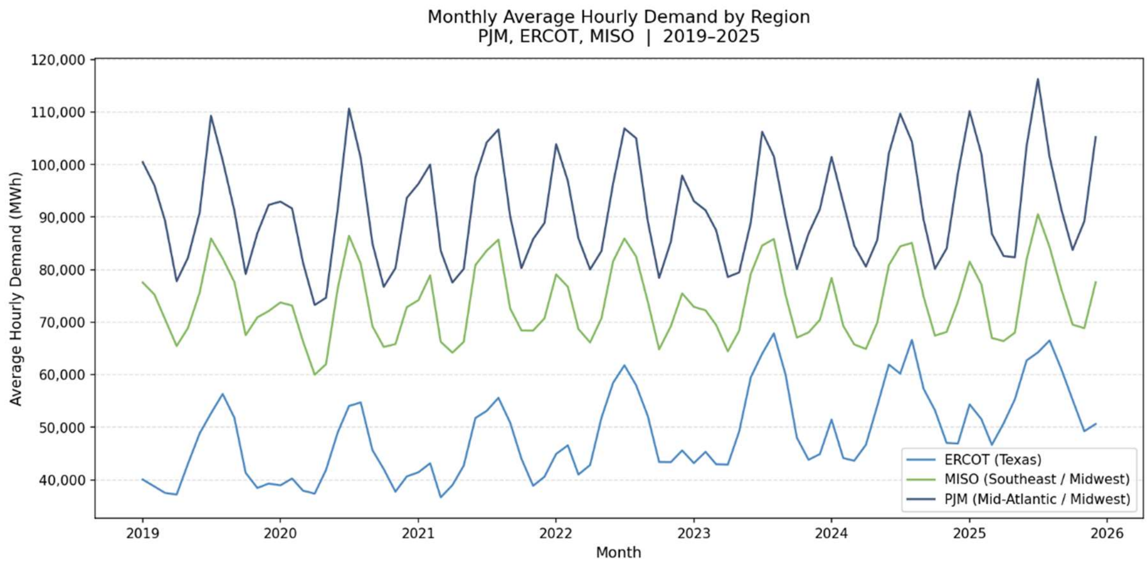


**Figure A4.** Monthly Average Hourly Electricity Demand by Region, 2019 — 2025.

*Notes: Raw megawatt-hour demand showing seasonal variation. The ERCOT series exhibits a rising floor: minimum monthly values trend upward even as seasonal peaks fluctuate. This pattern motivates the minimum demand synthetic control specification. Source: EIA Form 930.*

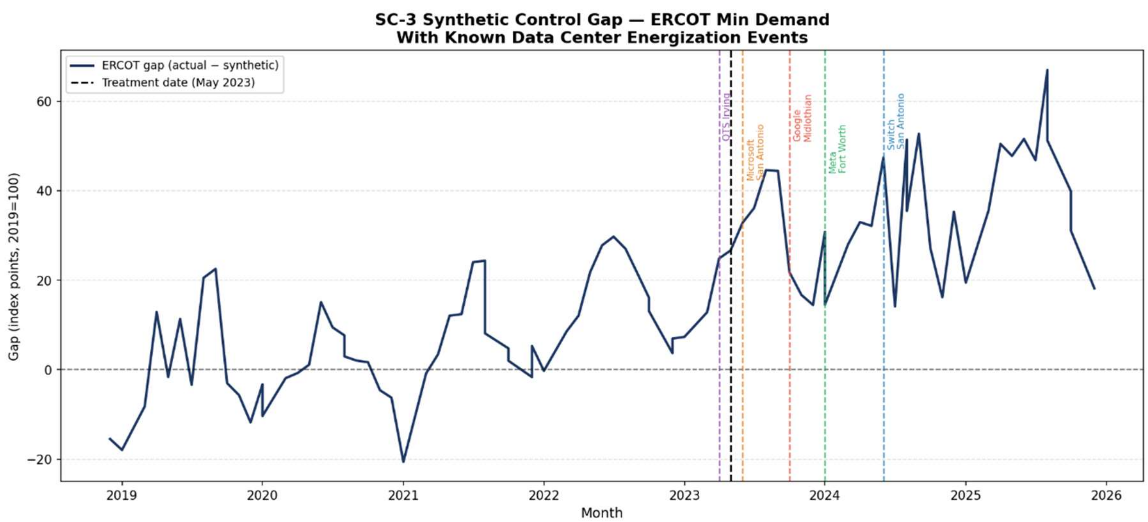


**Figure A5.** SC-3 Minimum Demand Gap with Individual Data Center Energization Markers.

*Notes: Gap plot (actual minus synthetic ERCOT minimum demand index) with separate colored dashed lines for each documented energization event. Companion detail to Figure 6 in the main text. Colored vertical lines: QTS Irving, Switch San Antonio, Meta Fort Worth, Microsoft San Antonio, Google Midlothian. Source: Authors' calculations; energization dates from public announcements.*

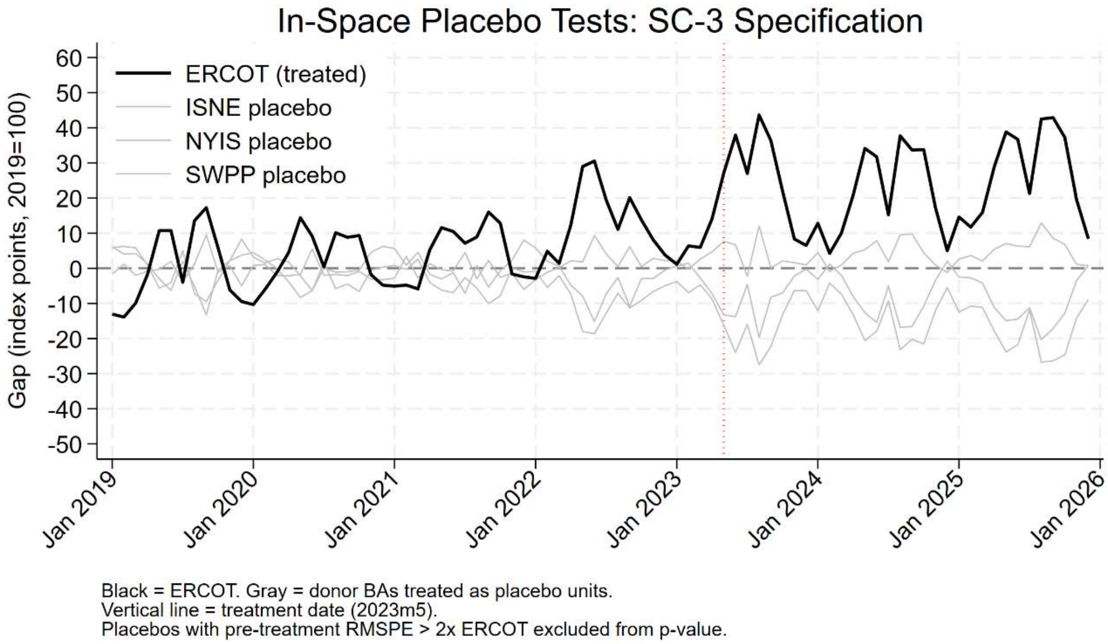


**Figure A6.** In-Space Placebo Tests — Minimum Demand SC-3 Specification.

*Notes: Black line is ERCOT; gray lines are donor BAs (ISNE, NYIS, SWPP) treated as placebo units. All placebos pass pre-treatment RMSPE filter (threshold = 27.06). Zero placebos exceed ERCOT post-treatment gap. Vertical dotted line: 2023m5 treatment date. Source: Authors' calculations.*

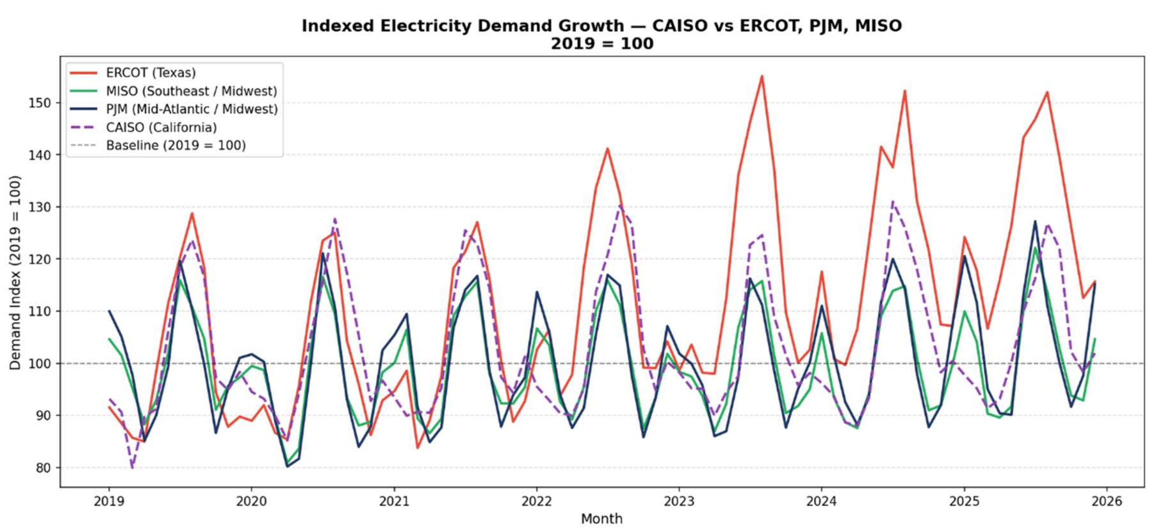


**Figure A7.** Indexed Electricity Demand Growth — ERCOT, PJM, MISO, and CAISO, 2019 — 2025.

*Notes: Monthly average hourly demand indexed to 2019 average = 100. CAISO (California, dashed) shown alongside ERCOT, PJM, and MISO for comparison. CAISO follows PJM and MISO rather than ERCOT, consistent with the null structural break result in Section 7. Source: EIA Form 930.*

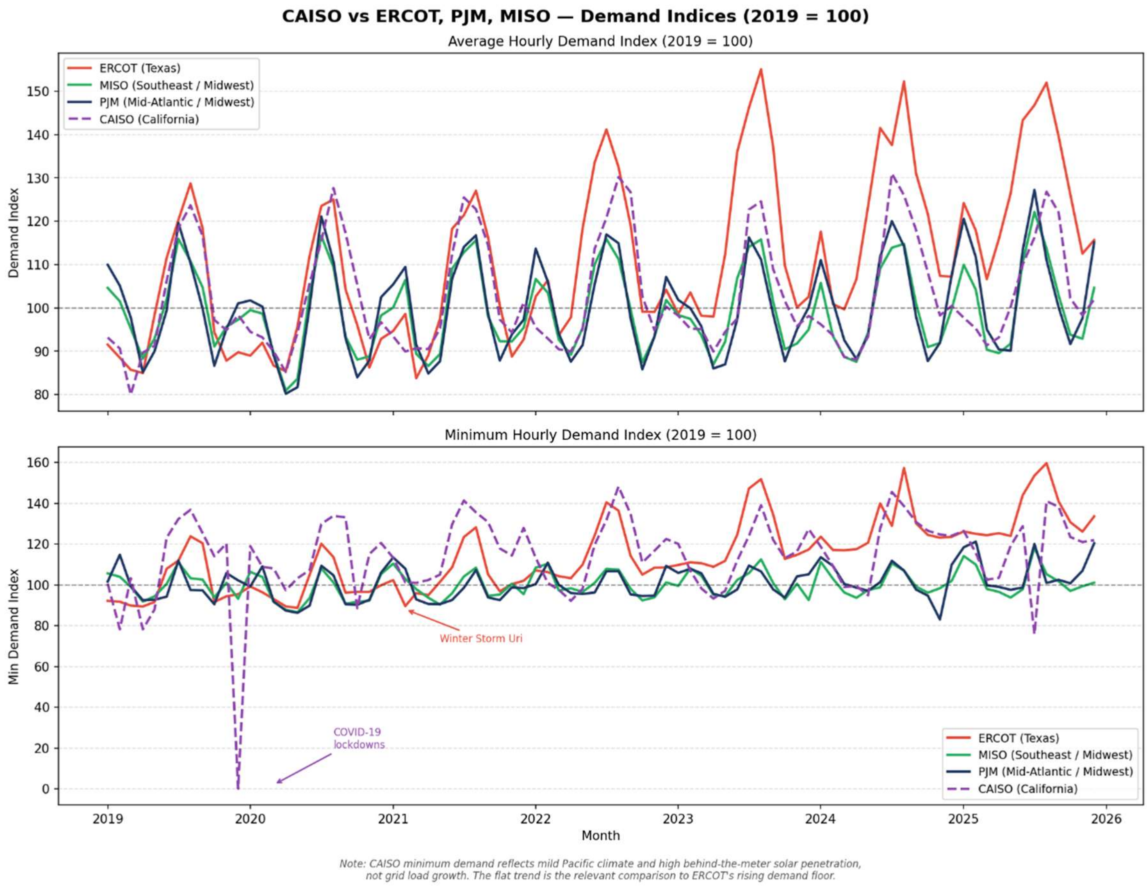


**Figure A8.** CAISO vs. ERCOT — Average and Minimum Hourly Demand Indices, 2019 — 2025.

*Notes: Two-panel figure showing both average (top) and minimum (bottom) hourly demand indices. The COVID-19 lockdown effect on CAISO minimum demand in early 2020 reflects behind-the-meter solar and commercial load dynamics. CAISO minimum demand is structurally distinct from ERCOT's: mild Pacific climate and high solar penetration create a different demand floor regime. Source: EIA Form 930.*

## Appendix B. Robustness and Sensitivity Figures

The following figures support the robustness of the synthetic control results and provide additional specification checks. Figures B1 — B6 correspond to the main ERCOT analysis in Section 6. Figures B7 — B9 correspond to the CAISO comparative analysis in Section 7.

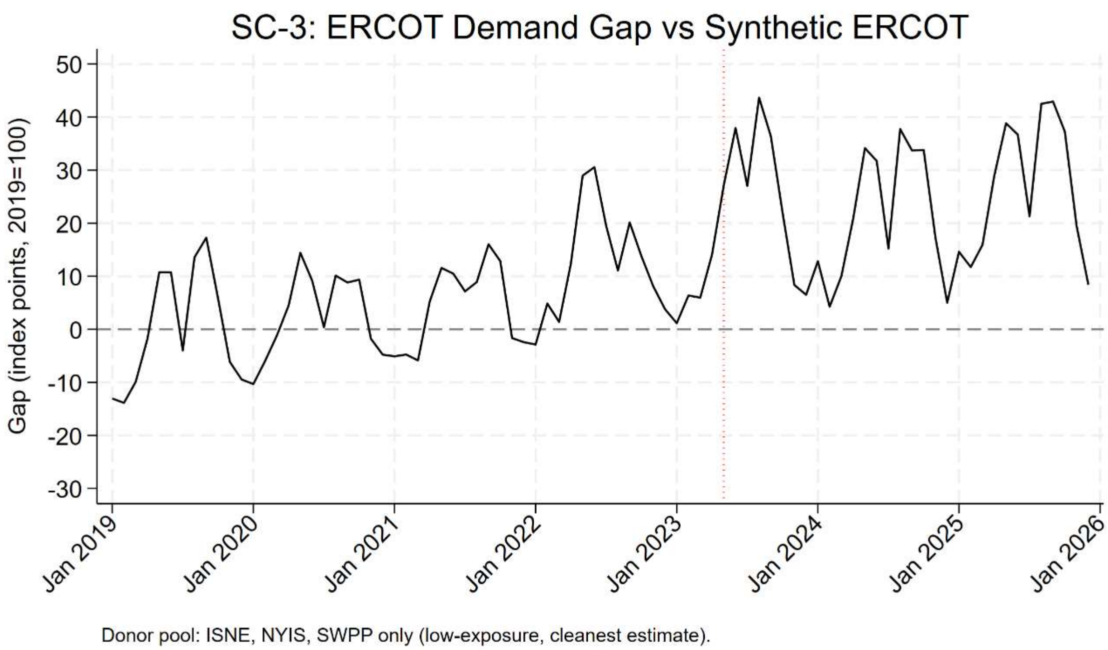


**Figure B1.** Synthetic Control Gap — ERCOT Average Hourly Demand, SC-3 Specification.

*Notes: Average demand SC-3 gap. Post-treatment mean gap: 24.5 index points. The smaller gap relative to the minimum demand specification (34.8 index points, Figure 6) is consistent with weather-driven load variation diluting the always-on infrastructure signal in the average demand series. Source: Authors' calculations.*

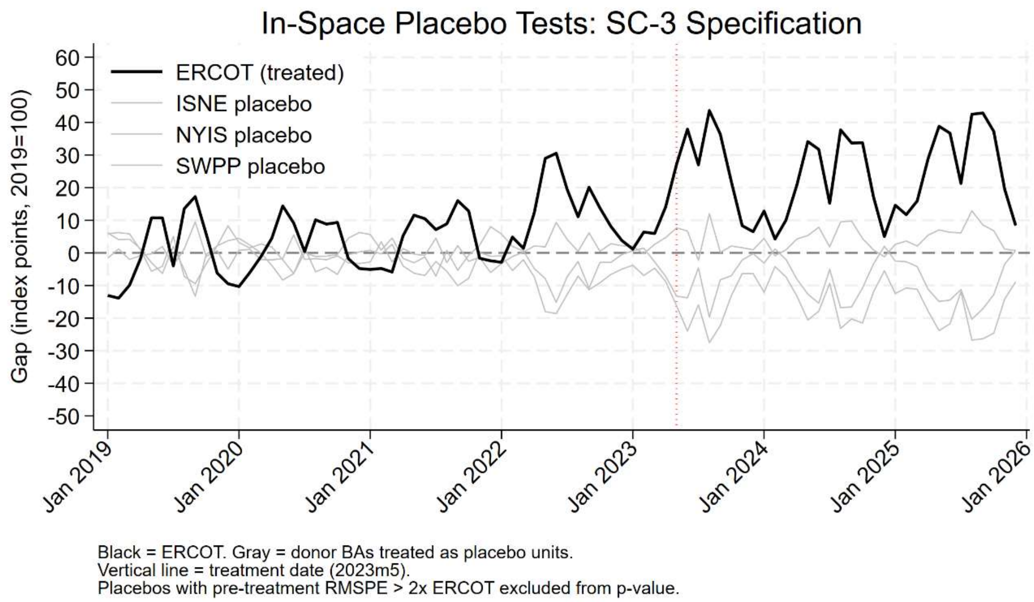


**Figure B2.** In-Space Placebo Tests — Average Demand SC-3 Specification.

*Notes: In-space placebo distribution for the average demand synthetic control. Zero placebos exceed ERCOT post-treatment gap (p = 0.25). Source: Authors' calculations.*

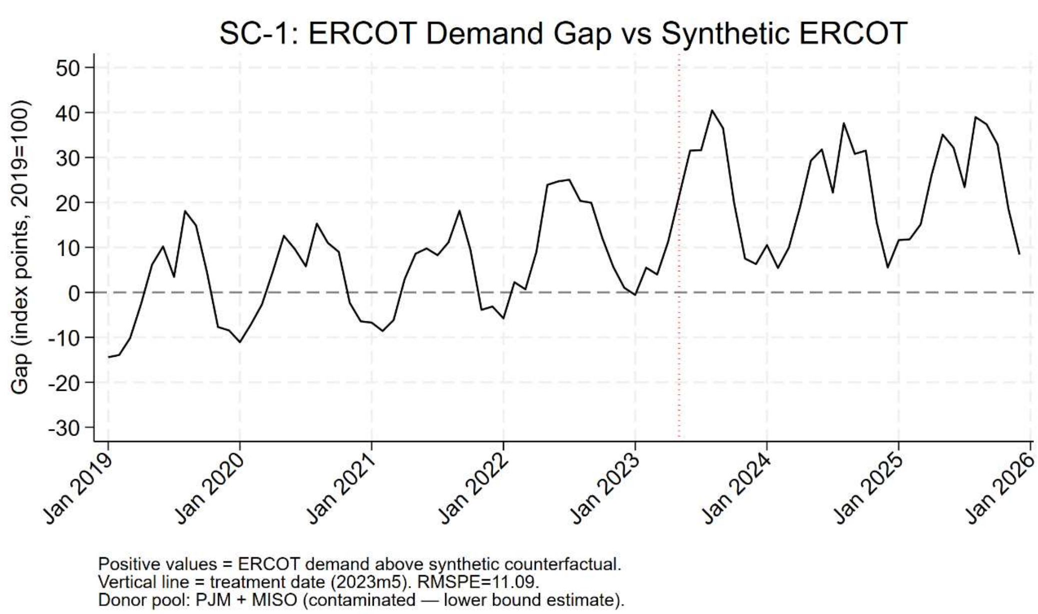


**Figure B3.** Synthetic Control Gap — SC-1 Specification (PJM and MISO Donors).

*Notes: Post-treatment mean gap: 23.0 index points. This contaminated specification provides the lower bound — PJM's footprint includes Northern Virginia, the largest data center market in the country. The SC-1 < SC-3 ordering confirms that including data-center-exposed donors suppresses the estimated gap. Pre-treatment RMSPE = 11.09. Source: Authors' calculations.*

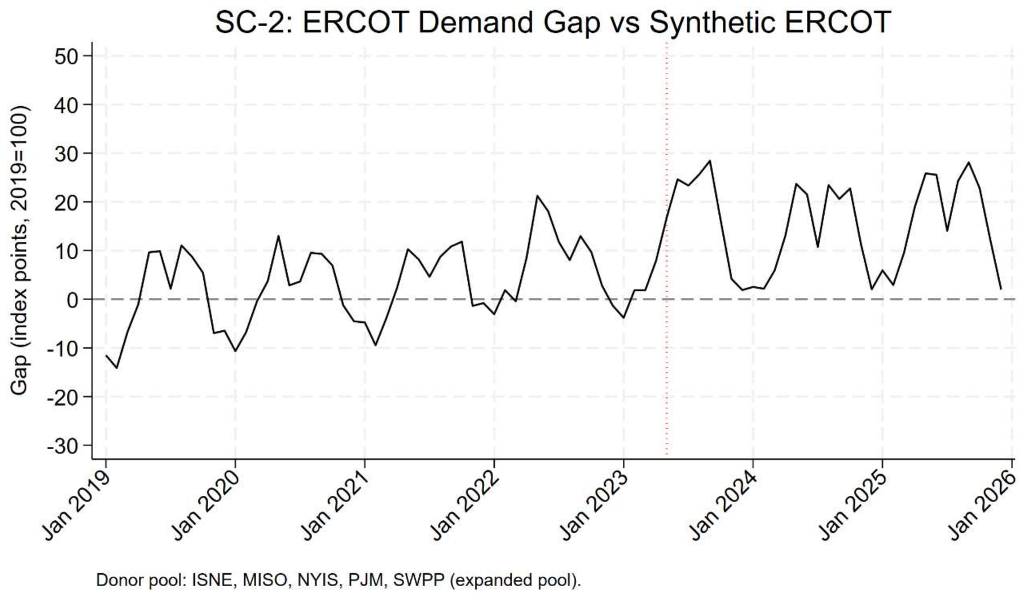


**Figure B4.** Synthetic Control Gap — SC-2 Specification (Expanded Donor Pool).

*Notes: SC-2 expands to all available balancing authorities above a coverage threshold. Post-treatment gap approximately 24.0 index points, between SC-1 and SC-3 as predicted by the contamination ordering hypothesis. Source: Authors' calculations.*

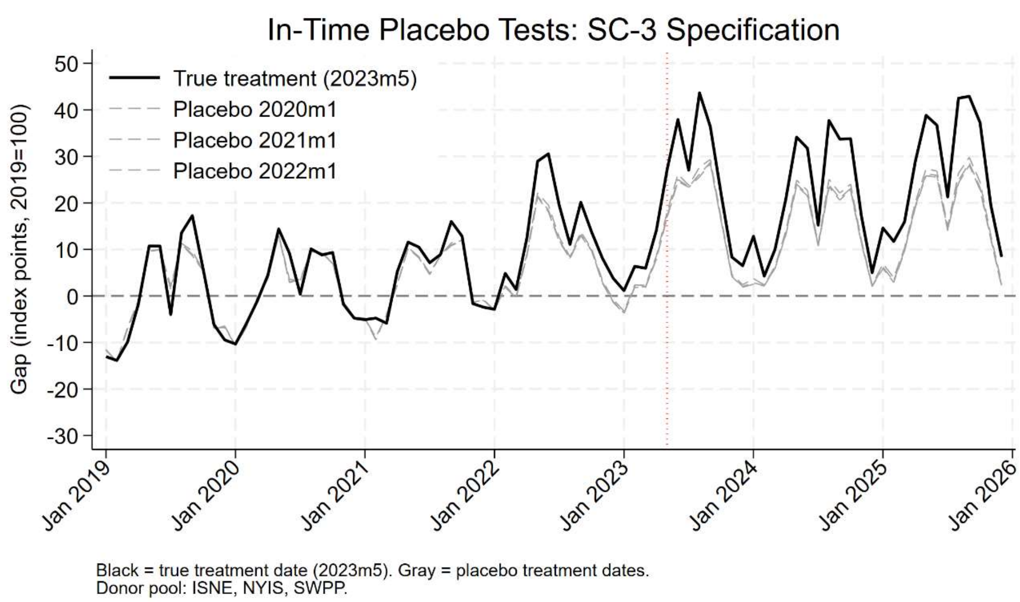


**Figure B5.** In-Time Placebo Test.

*Notes: Placebo treatment dates assigned at earlier points in the sample. The ERCOT gap associated with the true treatment date (2023m5) is larger than gaps at all placebo dates, providing additional inferential support. Source: Authors' calculations.*

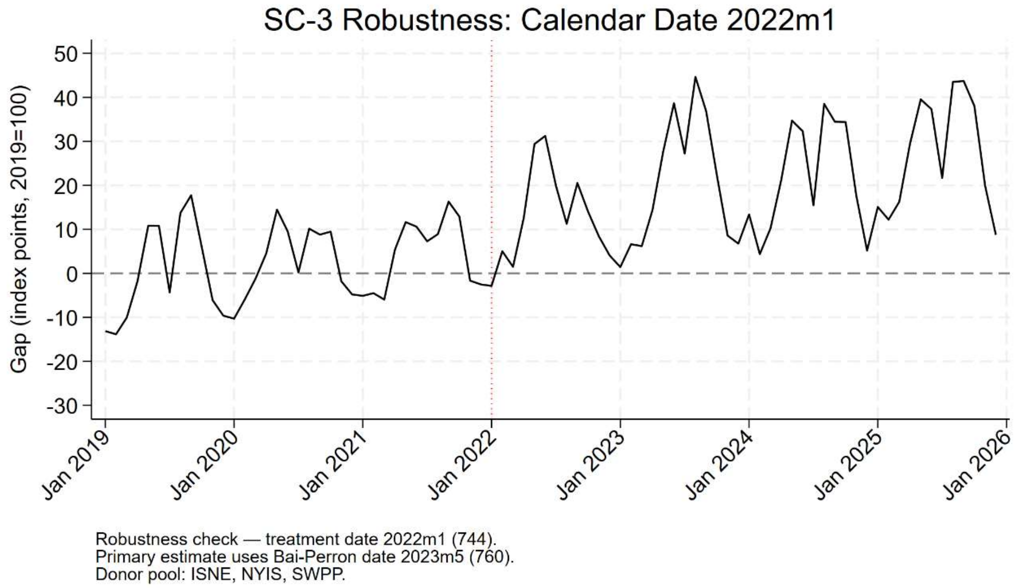


**Figure B6.** Synthetic Control Robustness — Alternative Treatment Date (January 2022).

*Notes: SC-3 minimum demand specification using an alternative treatment date of January 2022 rather than the data-determined Bai-Perron break of May 2023. The post-treatment mean gap under the 2022m1 date is 25.0 index points, compared to 34.8 index points under the primary 2023m5 specification. The smaller gap under the earlier date confirms 2023m5 as the correct structural break — facilities were still under construction in early 2022 and had not yet loaded onto the grid. Source: Authors' calculations.*

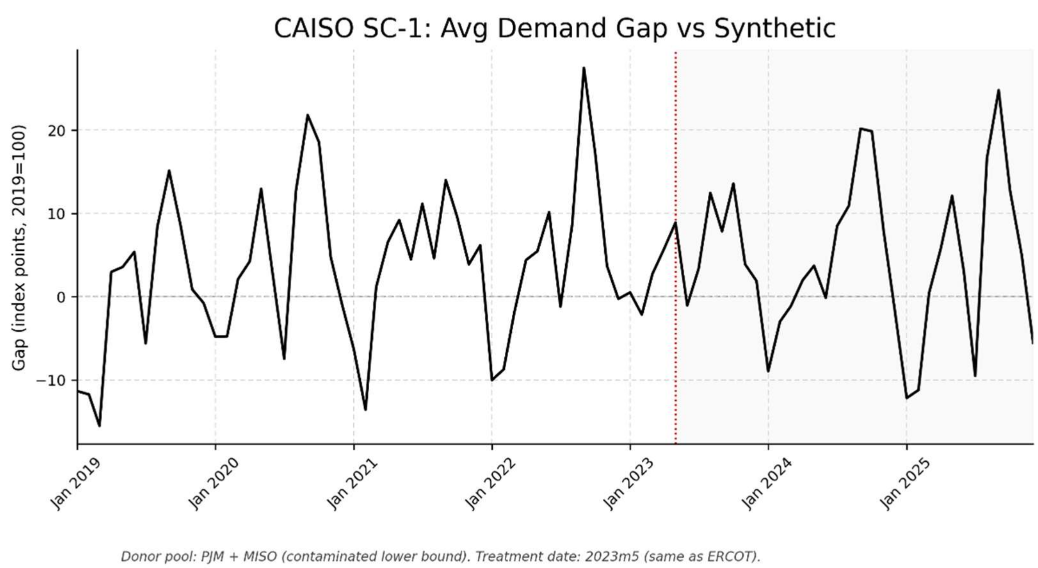


**Figure B7.** CAISO Synthetic Control — SC-1 Specification.

*Notes: CAISO SC-1 gap using PJM and MISO as donors. Post-treatment gap is small and variable, consistent with the null structural break result (UDmax = 4.00). Source: Authors' calculations.*

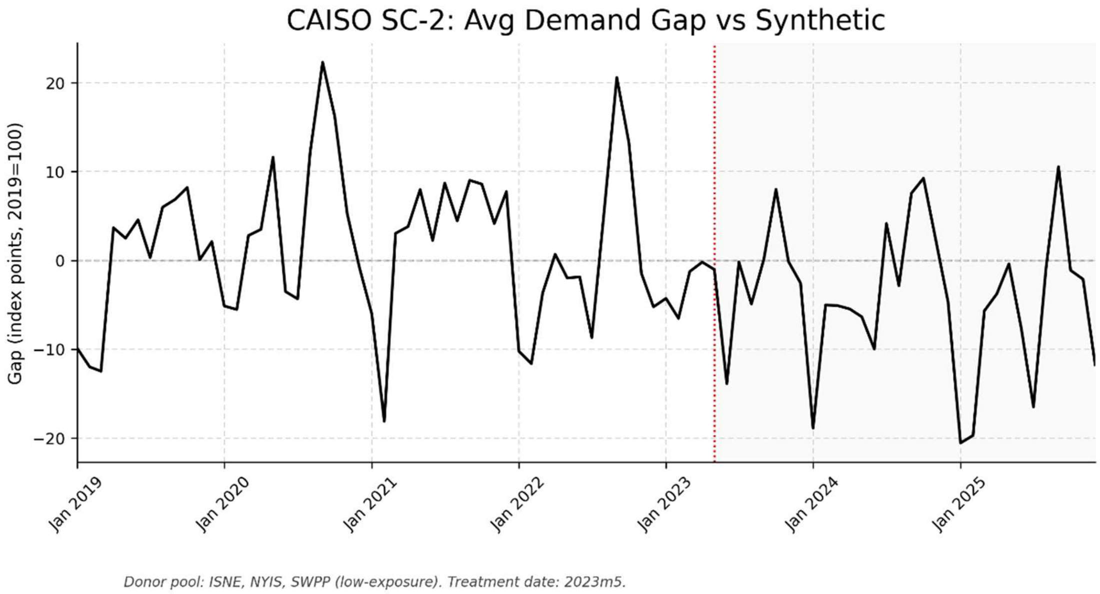


**Figure B8.** CAISO Synthetic Control — SC-2 Specification.

*Notes: CAISO SC-2 (expanded donor pool). Gap remains near zero post-treatment across specifications. Source: Authors' calculations.*

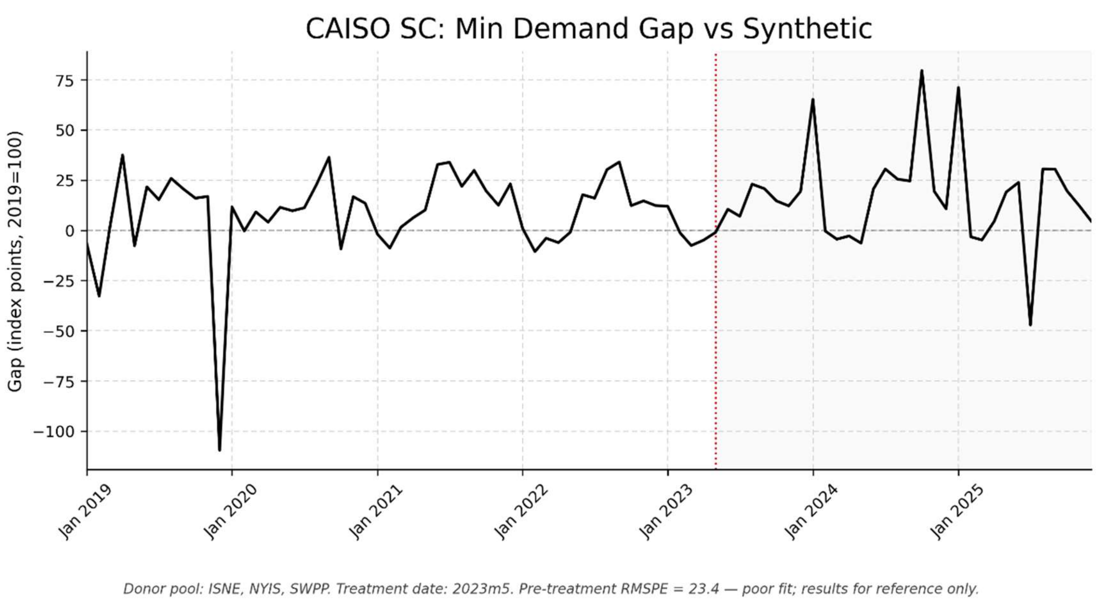


**Figure B9.** CAISO Synthetic Control — Minimum Demand Specification.

*Notes: Pre-treatment RMSPE = 23.4 — poor fit, reflecting CAISO-specific minimum demand dynamics (behind-the-meter solar, mild Pacific climate). Results reported for reference only; the high RMSPE precludes credible causal inference. Source: Authors' calculations.*

## Appendix C. Data Documentation

### C.1 Project Size Distribution

Figure C1 presents the distribution of project sizes in the LBNL interconnection queue data for PJM, ERCOT, and MISO. The distribution is smooth and right-skewed with no visible natural break, confirming that the 100 megawatt threshold was not selected by identifying a modal gap in the data. The 100 megawatt threshold corresponds to the industry definition of hyperscale infrastructure (BloombergNEF, 2025) and was determined before any regression results were examined.

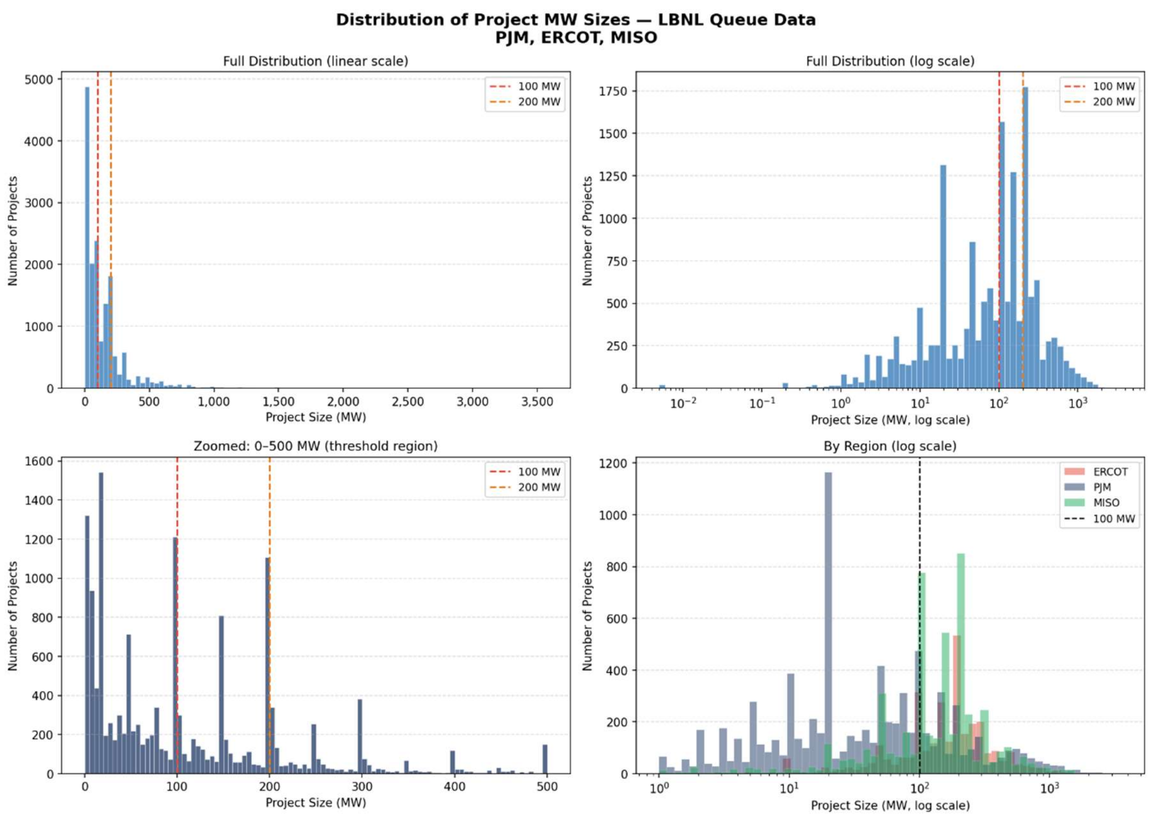


**Figure C1.** Distribution of Project MW Sizes — LBNL Queue Data, PJM, ERCOT, and MISO.

*Notes: Distribution of all project sizes in the LBNL Queued Up (2024 edition) data for PJM, ERCOT, and MISO. The distribution is smooth and right-skewed with no visible natural break, confirming the 100 MW threshold was not selected by identifying a modal gap in the data. The 100 MW threshold (dashed orange line) corresponds to the industry definition of hyperscale infrastructure (BloombergNEF, 2025) and was determined before any regression results were examined. Source: LBNL Queued Up 2024; authors' calculations.*

## C.2 Variable Definitions

| Variable | Definition |
| --- | --- |
| `avg_demand_mwh` | Average hourly electricity demand within a BA-month (MWh). Mean of all hourly EIA Form 930 readings within the BA-month. |
| `min_demand_mwh` | Minimum hourly electricity demand within a BA-month (MWh). Lowest single hourly reading; isolates the baseload floor invariant to weather, time of day, and season. |
| `queue_mw_filed_large` | Total megawatts of new generation interconnection requests filed within a BA-month, restricted to projects at or above 100 MW. Primary treatment variable. |
| `queue_mw_active` | Cumulative megawatts ever filed in a BA through a given month. Stock variable used in robustness checks. |
| `hdd` | Monthly heating degree days by balancing authority. Source: NOAA National Centers for Environmental Information. |
| `cdd` | Monthly cooling degree days by balancing authority. Source: NOAA National Centers for Environmental Information. |
| `gdp` | Annual real GDP by state (2017 dollars), interpolated monthly and assigned to balancing authorities by geographic crosswalk. Source: Bureau of Economic Analysis regional accounts. |

### C.3 Winsorization Note

The variable queue_mw_large_lag18 (queue_mw_filed_large lagged 18 months) is winsorized at its 95th percentile (18,588 MW) before all regressions. Two extreme observations in the raw distribution are traceable to the documented MISO 2022 queue reform backlog processing spike propagated 18 months forward: 154,214 MW (2024m3, MISO) and 102,870 MW (2025m10, MISO). Without winsorization, these observations create extreme leverage that drives the primary coefficient negative. The jump from the 90th percentile (9,350 MW) to the 99th percentile (102,870 MW) is a ten-fold gap with no observations in between, confirming these are institutional outliers rather than genuine data center activity. Nine observations are affected. The same p95 cap is applied consistently across all lag specifications in the robustness table

### C.4 CAISO Data Note

CAISO demand data were pulled from EIA Form 930 using respondent code CISO. CAISO minimum demand exhibits unusually high volatility in early 2020, traceable to COVID-19 lockdown effects on behind-the-meter solar generation and commercial load collapse. This structural feature contributes to the poor pre-treatment fit in the minimum demand synthetic control specification (RMSPE = 23.4) and is the primary reason the CAISO minimum demand SC results are reported as reference only rather than as credible causal estimates in Section 7.